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Device and method for determining a cardiac sensing control parameter

Abstract

A medical device processor is configured to receive a first cardiac electrical signal sensed from a first sensing electrode vector, receive a second cardiac electrical signal sensed from a second sensing electrode vector different than the first sensing electrode vector, and construct a third cardiac electrical signal from the first cardiac electrical signal and the second cardiac electrical signal. In some examples, the system determines sensed cardiac events according to at least one setting of a cardiac event sensing threshold control parameter from at least the third cardiac electrical signal and may determine at least one acceptable setting of a sensing control parameter based on the determined sensed cardiac events. The processor may generate an output representative of the determined sensed cardiac events.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. patent application Ser. No. 17/350,153, filed Jun. 17, 2021, which claims the benefit of provisional U.S. Patent Application No. 63/145,687, filed on Feb. 4, 2021, and provisional U.S. Patent Application No. 63/045,567, filed on Jun. 29, 2020, all of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The disclosure relates generally to a medical device and method for determining a sensing control parameter to promote reliable sensing of cardiac event signals.

BACKGROUND

(2) Medical devices may sense electrophysiological signals from the heart, brain, nerve, muscle or other tissue. Such devices may be implantable, wearable or external devices using implantable and/or surface (skin) electrodes for sensing the electrophysiological signals. In some cases, such devices may be configured to deliver a therapy based on the sensed electrophysiological signals. For example, implantable or external cardiac pacemakers, cardioverter defibrillators, cardiac monitors and the like, sense cardiac electrical signals from a patient's heart. A cardiac pacemaker or cardioverter defibrillator may sense cardiac electrical signals from the heart and deliver electrical stimulation therapies to the heart using electrodes carried by a transvenous medical electrical lead, a non-transvenous medical electrical lead and/or leadless electrodes coupled directly to the housing of the medical device.

(3) The electrical stimulation therapies may include signals such as pacing pulses or cardioversion/defibrillation shocks. In some cases, a medical device may sense cardiac event signals attendant to the intrinsic or pacing-evoked depolarizations of the heart and control delivery of stimulation signals to the heart based on sensed cardiac event signals. Upon detection of an abnormal rhythm based on the sensed cardiac event signals (or absence thereof), such as bradycardia, tachycardia or fibrillation, an appropriate electrical stimulation signal or signals may be delivered to restore or maintain a more normal rhythm of the heart. For example, an implantable cardioverter defibrillator (ICD) may deliver pacing pulses to the heart of the patient upon detecting bradycardia or tachycardia or deliver cardioversion/defibrillation (CV/DF) shocks to the heart upon detecting tachycardia or fibrillation. Reliable sensing of the cardiac event signals attendant to myocardial depolarizations from cardiac electrical signals is important in controlling appropriate electrical stimulation therapies for the benefit of the patient.

SUMMARY

(4) In general, the disclosure is directed to a device and method for determining sensed cardiac events from a cardiac electrical signal for use in identifying an acceptable or recommended setting of a sensing control parameter. The sensing control parameter may be, for example, a programmable sensing electrode vector or a cardiac event sensing threshold control parameter used in setting the amplitude of the sensing threshold used to sense cardiac event signals. A medical device system configured to sense cardiac electrical signals may include multiple available sensing electrode vectors, each vector being a different combination of electrodes selected from at least three electrodes, for sensing cardiac electrical signals. A medical device system operating according to the techniques disclosed herein senses two cardiac electrical signals using two different sensing electrode vectors and constructs at least one cardiac electrical signal from the two sensed cardiac electrical signals. The medical device system may determine sensed cardiac event signals from

multiple cardiac electrical signals, sensed and constructed, according to one or more settings of a cardiac event sensing threshold control parameter for use in determining an acceptable or recommended setting of a sensing control parameter.

(5) In one example, the disclosure provides a medical device comprising a processor configured to receive at least a first sensed cardiac electrical signal and a second sensed cardiac electrical signal and construct a third cardiac electrical signal from the first sensed cardiac electrical signal and the second sensed cardiac electrical signal. The processor determines sensed cardiac events according to at least one setting of a cardiac event sensing threshold control parameter from at least the third cardiac electrical signal and generates an output representative of the determined sensed cardiac events.

(6) In another example, the disclosure provides a method that includes receiving at least a first sensed cardiac electrical signal and a second sensed cardiac electrical signal and constructing a third cardiac electrical signal from the first sensed cardiac electrical signal and the second sensed cardiac electrical signal. The method further includes determining sensed cardiac events from at least the third cardiac electrical signal according to at least one setting of a cardiac event sensing threshold control parameter and generating an output representative of the determined sensed cardiac events.

(7) In another example, the disclosure provides a non-transitory computer-readable medium storing a set of instructions which, when executed by processing circuitry of a medical device system, cause the processing circuitry to receive at least a first sensed cardiac electrical signal and a second sensed cardiac electrical signal and construct a third cardiac electrical signal from the first sensed cardiac electrical signal and the second sensed cardiac electrical signal. The instructions further cause the processing circuitry to determine sensed cardiac events from at least the third cardiac electrical signal according to at least one setting of a cardiac event sensing threshold control parameter and generate an output representative of the determined sensed cardiac events.

(8) In another example, the disclosure provides a graphical user interface system including a processor configured to receive a cardiac electrical signal, determine sensed cardiac events from the cardiac electrical signal according to at least one setting of a sensing threshold control parameter and generate an output of data representative of the determined sensed cardiac events. The graphical user interface system further includes a display unit coupled to the processor and configured to receive the generated output of data from the processor and display a visual representation of the data representative of the determined sensed cardiac events.

(9) Further disclosed herein is the subject matter of the following clauses:

(10) 1. A medical device comprising a processor configured to:

(11) receive at least a first sensed cardiac electrical signal and a second sensed cardiac electrical signal; construct a third cardiac electrical signal from the first sensed cardiac electrical signal and the second sensed cardiac electrical signal; from at least the third cardiac electrical signal, determine sensed cardiac events according to at least one setting of a cardiac event sensing threshold control parameter; and generate an output representative of the determined sensed cardiac events.

2. The device of clause 1 wherein: the device includes a display unit; the processor is configured to: determine an acceptable setting of at least one of a sensing electrode vector and a cardiac event sensing threshold control parameter; and generate the output representative of the determined sensed cardiac events comprising the acceptable setting; the display unit being configured to display the acceptable setting.

3. The device of any of clauses 1-2, wherein the processor is configured to determine the sensed cardiac events according to a plurality of settings of the cardiac event sensing threshold control parameter by adjusting at least one of a sensitivity, a starting amplitude of a cardiac event sensing threshold amplitude, an intermediate amplitude of the cardiac event sensing threshold amplitude, or a time interval used for timing an adjustment to the cardiac event sensing threshold.

4. The device of any of clauses 1-3, wherein the processor is configured to determine at least one acceptable sensing control parameter setting based on the determined sensed cardiac events by: determining a rate of the determined sensed cardiac events; determining that the rate meets expected rate criteria; and determining that an associated sensing control parameter used to determine the sensed cardiac events is an acceptable sensing control parameter in response to the rate meeting the expected rate criteria.
5. The device of any of clauses 1-4, wherein the processor is configured to: detect a tachyarrhythmia based on the determined sensed cardiac events from at least one of the first cardiac electrical signal, the second cardiac electrical signal or the third cardiac electrical signal; and determine at least one acceptable setting of a sensing control parameter by identifying a sensing control parameter associated with the detected tachyarrhythmia.
6. The device of clause 5, wherein the processor is configured to: determine a time to detect the tachyarrhythmia for each of a plurality of settings of the cardiac event sensing threshold control parameter for a given one of the first cardiac electrical signal, second cardiac electrical signal and third cardiac electrical signal; and determine the at least one acceptable setting of the sensing control parameter from among the plurality of settings of the cardiac event sensing threshold control parameter associated with a determined time to detect the tachyarrhythmia that is within a tachyarrhythmia detection time limit.
7. The device of clause 6, wherein the processor is further configured to determine the tachyarrhythmia detection time limit by: determining a minimum time to detect the tachyarrhythmia among the times to detect the tachyarrhythmia determined for the plurality of settings of the sensing threshold control parameter; and determining the tachyarrhythmia detection time limit as the minimum time plus a predetermined increase in the time to detect the tachyarrhythmia.
8. The device of any of clauses 6-7, wherein the processor is further configured to determine the at least one acceptable sensing control parameter setting by: determining a highest value of a sensitivity setting associated with a time to detect the tachyarrhythmia that is less than or equal to the tachyarrhythmia detection time limit; determine an acceptable sensitivity setting that is a factor of the highest value, wherein the factor corresponds to a predetermined safety margin for sensing cardiac event signals.
9. The device of any of clauses 1-8, comprising: a telemetry unit configured to receive the first cardiac electrical signal and the second cardiac electrical signal transmitted from another medical device; wherein the processor is configured to: receive the first cardiac electrical signal and the second cardiac electrical signal from the telemetry unit, where the first cardiac electrical signal corresponds to a first sensing electrode vector comprising a first electrode and a second electrode, the second cardiac electrical signal corresponds to a second sensing electrode vector comprising the first electrode and a third electrode, and construct the third cardiac electrical signal corresponding to a third sensing electrode vector including the second electrode and the third electrode.
10. The device of any of clauses 1-9, comprising: a sensing circuit configured to: sense the first cardiac electrical signal from a first sensing electrode vector comprising a first electrode and a second electrode; sense a second cardiac electrical signal from a second sensing electrode vector comprising the first electrode and a third electrode; and the processor is configured to receive the first cardiac electrical signal and the second cardiac electrical signal from the sensing circuit and construct the third cardiac electrical signal corresponding to a third sensing electrode vector including the second electrode and the third electrode.
11. The device of any of clauses 1-9 further comprising a display unit, wherein the processor is configured to generate the output by generating data corresponding to the determined sensed cardiac events for display by the display unit in a graphical user interface.
12. The device of clause 11, wherein the processor is configured to receive a user input indicating the at least one setting of the cardiac event sensing threshold control parameter.

13. The device of any of clauses 11-12, wherein the processor is configured to: receive a user input indicating a sensing electrode vector; and construct the third cardiac electrical signal corresponding to the sensing electrode vector.
14. The device of any of clauses 11-13, wherein the processor is configured to: receive a user input indicating a selection of at least one sensing control parameter setting from the displayed graphical user interface; and generate the output comprising a programming command corresponding to the user input indicating the selection of the at least one sensing control parameter.
15. The device of clause 14, wherein the processor is configured to receive the user input indicating the selection of at least one sensing control parameter setting from the displayed graphical user interface as a combination of a sensing electrode vector and a cardiac event sensing threshold control parameter.
16. The device of any of clauses 2-15, further comprising a telemetry unit configured to transmit a programming command including the acceptable sensing control parameter setting.
17. A method comprising: receiving at least a first sensed cardiac electrical signal and a second sensed cardiac electrical signal; constructing a third cardiac electrical signal from the first sensed cardiac electrical signal and the second sensed cardiac electrical signal; from at least the third cardiac electrical signal, determining sensed cardiac events according to at least one setting of a cardiac event sensing threshold control parameter; and generating an output representative of the determined sensed cardiac events.
18. The method of clause 17, comprising: determining an acceptable setting of at least one of a sensing electrode vector and a cardiac event sensing threshold control parameter; generating the output representative of the determined sensed cardiac events comprising the acceptable setting; and displaying the acceptable setting by a display unit.
19. The method of any of clauses 17-18, comprising determining the sensed cardiac events according to a plurality of settings of the cardiac event sensing threshold control parameter by adjusting at least one of a sensitivity, a starting amplitude of a cardiac event sensing threshold amplitude, an intermediate amplitude of the cardiac event sensing threshold amplitude, or a time interval used for timing an adjustment to the cardiac event sensing threshold amplitude.
20. The method of any of clauses 17-19, further comprising determining at least one acceptable sensing control parameter setting based on the determined sensed cardiac events by: determining a rate of the determined sensed cardiac events; determining that the rate meets expected rate criteria; and determining that an associated sensing control parameter used to determine the sensed cardiac events is the acceptable sensing control parameter in response to the rate meeting the expected rate criteria.
21. The method of any of clauses 17-20, further comprising detecting a tachyarrhythmia based on the determined sensed cardiac events from at least one of the first cardiac electrical signal, the second cardiac electrical signal or the third cardiac electrical signal; determining at least one acceptable setting of a sensing control parameter by identifying a sensing control parameter associated with the detected tachyarrhythmia.
22. The method of clause 21, comprising: determining a time to detect the tachyarrhythmia for each of a plurality of settings of the cardiac event sensing threshold control parameter for a given one of the first cardiac electrical signal, second cardiac electrical signal and third cardiac electrical signal; and determining the at least one acceptable setting of the sensing control parameter by identifying one of the plurality of settings of the cardiac event sensing threshold control parameter associated with a time to detect the tachyarrhythmia that is within a tachyarrhythmia detection time limit.
23. The method of clause 22, wherein determining the tachyarrhythmia detection time limit comprises: determining a minimum time to detect the tachyarrhythmia among the times to detect the tachyarrhythmia determined for the plurality of settings of the sensing threshold control parameter; and determining the tachyarrhythmia detection time limit as the minimum time plus a predetermined increase in the time to detect the tachyarrhythmia.

24. The method of any of clauses 22-23, wherein determining the at least one acceptable setting of the sensing control parameter comprises: determining a highest value of a sensitivity setting associated with a time to detect the tachyarrhythmia that is less than or equal to the tachyarrhythmia detection time limit; determining the acceptable setting of the sensing control parameter as a sensitivity setting that is a factor of the highest value, wherein the factor corresponds to a predetermined safety margin for sensing cardiac event signals.

25. The method of any of clauses 17-24, comprising: receiving the first cardiac electrical signal and the second cardiac electrical signal transmitted from another medical device, where the first cardiac electrical signal corresponds to a first sensing electrode vector comprising a first electrode and a second electrode, the second cardiac electrical signal corresponds to a second sensing electrode vector comprising the first electrode and a third electrode, and constructing the third cardiac electrical signal corresponding to a third sensing electrode vector including the second electrode and the third electrode.

26. The method of any of clauses, 17-25, comprising: sensing the first cardiac electrical signal from a first sensing electrode vector comprising a first electrode and a second electrode; sensing a second cardiac electrical signal from a second sensing electrode vector comprising the first electrode and a third electrode; and constructing the third cardiac electrical signal corresponding to a third sensing electrode vector including the second electrode and the third electrode.

27. The method of any of clauses 17-26, wherein generating the output comprises: generating data corresponding to the determined sensed cardiac events; and displaying the generated data in a graphical user interface.

28. The method of clause 27, comprising receiving a user input, the user input indicating the at least one setting of the cardiac event sensing threshold control parameter.

29. The method of any of clauses 27-28, comprising: receiving a user input indicating a sensing electrode vector; and constructing the third cardiac electrical signal corresponding the sensing electrode vector.

30. The method of any of clauses 27-29, comprising: receiving a user input indicating a selection of at least one sensing control parameter setting from the displayed graphical user interface; wherein generating the output comprises generating a programming command corresponding to the user input indicating the selection of the at least one sensing control parameter.

31. The method of clause 27, comprising receiving the user input indicating the selection of at least one sensing control parameter setting from the display as a combination of a sensing electrode vector and a cardiac event sensing threshold control parameter.

32. The method of any of clauses 18-31, comprising transmitting a programming command including the acceptable sensing control parameter setting.

33. A non-transitory computer-readable medium storing a set of instructions which, when executed by a processor of a medical device, cause the device to: receive at least a first sensed cardiac electrical signal and a second sensed cardiac electrical signal; construct a third cardiac electrical signal from the first sensed cardiac electrical signal and the second sensed cardiac electrical signal; from at least the third cardiac electrical signal, determine sensed cardiac events according to at least one setting of a cardiac event sensing threshold control parameter; and generate an output representative of the determined sensed cardiac events.

34. A graphical user interface system comprising: a processor configured to: obtain a plurality of cardiac electrical signals sensed via a plurality of sensing electrode vectors; predict, for each of a plurality of cardiac electrical signals, whether tachyarrhythmia detection is expected to be made from the corresponding cardiac electrical signal at a plurality of sensitivity settings; determine one or more acceptable sensitivity settings of at least one of the plurality of sensing electrode vectors based on at least the predictions; and generate an output of data representative of acceptable sensitivity settings for tachyarrhythmia detection for the respective plurality of sensing electrode vectors; and a display unit coupled to the processor and configured to: receive the generated output

of data from the processor; and display a visual representation of the acceptable sensitivity settings for tachyarrhythmia detection for the respective plurality of sensing electrode vectors.

35. The graphical user interface system of clause 34, wherein the processor is configured to: determine one or more acceptable sensitivity settings of at least one of plurality of sensing electrode vectors by determining a recommended sensitivity setting for one of the plurality of sensing electrode vectors based on at least the predictions; and generate the output of data to include the recommended sensitivity setting.

(12) This summary is intended to provide an overview of the subject matter described in this disclosure. It is not intended to provide an exclusive or exhaustive explanation of the apparatus and methods described in detail within the accompanying drawings and description below. Further details of one or more examples are set forth in the accompanying drawings and the description below.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIGS. 1A and 1B are conceptual diagrams of a medical device system configured to sense cardiac electrical signals and deliver cardiac electrical stimulation therapies according to one example.

(2) FIG. 1C is a conceptual diagram of a medical device system including an external cardiac monitor capable of wireless communication with an ICD for receiving cardiac electrical signals sensed by the ICD and/or sensing surface electrocardiogram signals.

(3) FIGS. 2A-2C are conceptual diagrams of a patient implanted with the medical device system of FIGS. 1A and 1B in a different implant configuration.

(4) FIG. 3 is a conceptual diagram of a medical device system according to another example.

(5) FIG. 4 is a conceptual diagram of a medical device configured to sense cardiac electrical signals according to one example.

(6) FIG. 5 is a conceptual diagram of circuitry that may be included in the sensing circuit shown in FIG. 4 according to one example.

(7) FIG. 6 is a diagram of a filtered and rectified cardiac electrical signal illustrating one technique for adjusting a cardiac event sensing threshold.

(8) FIG. 7 is a flow chart of a method that may be performed by a medical device for determining a recommended setting of a sensing control parameter based on an analysis of multiple sensing electrode vector signals.

(9) FIG. 8 is a diagram of some possible sensing electrode vectors that exist between the electrodes carried by the lead shown in FIG. 1A and the ICD housing.

(10) FIG. 9 is a flow chart of a method performed by a medical device for determining sensed cardiac events from multiple sensing electrode vector signals according to another example.

(11) FIG. 10 is a diagram of three cardiac electrical signals that may be analyzed according to the techniques disclosed herein.

(12) FIG. 11 is a diagram of a graphical user interface (GUI) that may be generated as output by a processor for display on a display unit according to one example.

(13) FIG. 12 is a graph of tachyarrhythmia detection time data that may be determined by a processor of a medical device system and generated as output for display according to one example.

(14) FIG. 13 is a flow chart of a method that may be performed by a medical device system processor for determining a recommended setting of a sensing control parameter according to one example.

(15) FIG. 14 is a GUI that may be generated for display on a display unit of a medical device

according to one example.

(16) FIG. 15 is a diagram of a GUI that may be generated as output by a processor for display on a display unit of a device performing the disclosed techniques according to another example.

(17) FIG. 16 is another example of GUI that may be displayed by a medical device to present a visualization of sensed cardiac event related data determined by a processor.

(18) FIG. 17 is a flow chart of a method for determining acceptable or recommended sensing control parameters according to another example.

DETAILED DESCRIPTION

(19) In general, this disclosure describes a device and techniques for determining sensed cardiac events according to different sensing control parameters. Cardiac events may be sensed by a medical device according to programmed sensing control parameters for determining a cardiac rate and for detecting an abnormal cardiac rhythm, e.g., bradycardia, tachycardia, asystole, etc., for controlling the delivery of cardiac electrical stimulation therapy as needed. The “cardiac event” being sensed is an event associated with a single cardiac cycle such as an R-wave attendant to ventricular depolarization or a P-wave attendant to atrial depolarization. A cardiac event is determined to be sensed when a cardiac electrical signal crosses a cardiac event sensing threshold. Determining sensed cardiac events is therefore equivalent to determining a cardiac event sensing threshold crossing time by a cardiac electrical signal in some examples. Sensing control parameters used to determine sensed cardiac events from a cardiac electrical signal include the sensing electrode vector used to sense the cardiac electrical signal and cardiac event sensing threshold control parameters used to set the amplitude of the cardiac event sensing threshold at any given time during a cardiac cycle.

(20) Cardiac events may be sensed for detecting cardiac arrhythmias. For example, the rate of sensed cardiac events may be determined by a medical device for detecting atrial or ventricular tachyarrhythmia, such as atrial tachycardia (AT), atrial fibrillation (AF), ventricular tachycardia (VT) or ventricular fibrillation (VF). In other examples, the rate of sensed cardiac events may be determined for controlling cardiac pacing for treating bradycardia, asystole, long ventricular pause or other abnormal rhythms or conduction abnormalities. In some examples, one or more alternative settings of a cardiac event sensing threshold control parameter is used for determining cardiac events that are sensed for a given cardiac electrical signal. The cardiac electrical signal may be sensed using electrodes in a selected sensing electrode vector or may be a constructed cardiac electrical signal that is derived from two sensed cardiac electrical signals. One or more cardiac event sensing threshold control parameter settings may be applied to one or more sensed and/or constructed cardiac electrical signals during real time and/or during post processing signal analysis to determine sensed cardiac events. The determined sensed cardiac events may be used to predict cardiac event intervals and/or a cardiac event rate that would be determined from the cardiac electrical signal based on the applied sensing threshold control parameters.

(21) In some examples, the determination of cardiac event rates or intervals according to different sensing control parameter settings may be used to determine whether an arrhythmia detection would be detected and/or a predicted time of arrhythmia detection and/or therapy delivery. For instance, this determination may be used to determine a time required to detect a tachyarrhythmia episode. Using the predicted tachyarrhythmia detections and or predicted times to detect a tachyarrhythmia episode from a cardiac electrical signal according to multiple cardiac event sensing threshold control parameter settings, a medical device operating according to the techniques disclosed herein may determine acceptable settings of at least one sensing control parameter, such as the sensing electrode vector, sensitivity, or other cardiac event sensing threshold control parameters used to set the cardiac event sensing threshold amplitude during a cardiac cycle. The acceptable settings are likely to promote reliable sensing of cardiac events for detection of tachyarrhythmia and promote appropriate and timely therapy delivery as needed.

(22) The techniques disclosed herein for determining sensed cardiac events and analyzing the rate

and/or intervals of sensed cardiac events according to different sensing control parameter settings may be implemented in a device associated with a variety of cardiac device systems, such as a system including a cardiac monitor, pacemaker or ICD configured for sensing cardiac events and determining a cardiac event interval or rate for detecting a cardiac rhythm and, in some cases, controlling a cardiac electrical stimulation therapy. At least a portion of the techniques disclosed herein may be performed by a processor of an implantable medical device, such as a cardiac monitor, pacemaker or ICD, or by a processor of an external device configured to sense cardiac electrical signals or receive sensed cardiac electrical signals from another device.

(23) In the illustrative examples presented herein, a pacemaker or ICD is configured to sense cardiac electrical signals and deliver cardiac electrical stimulation pulses for pacing and/or CV/DF therapy delivery. The pacemaker or ICD may be coupled to a transvenous or non-transvenous lead in various examples for carrying electrodes for sensing cardiac electrical signals and delivering electrical stimulation therapy. For example, the pacemaker or ICD may be coupled to an “extra-cardiovascular” lead, referring to a lead that positions electrodes outside the blood vessels, heart, and pericardium surrounding the heart of a patient. Implantable electrodes carried by extra-cardiovascular leads, for example, may be positioned extra-thoracically (outside the ribcage and sternum) or intra-thoracically (beneath the ribcage or sternum, sometimes referred to as a sub-sternal position) but may not necessarily be in intimate contact with myocardial tissue. An extra-cardiovascular lead may also be referred to as a “non-transvenous” lead.

(24) In other examples, the medical device may be coupled to a transvenous lead that positions electrodes within a blood vessel, which may remain outside the heart in an “extra-cardiac” location or be advanced to position electrodes within a heart chamber. For instance, a transvenous medical lead may be advanced along a venous pathway to position electrodes in an extra-cardiac location within the internal thoracic vein (ITV), an intercostal vein, the superior epigastric vein, or the azygos, hemiazygos, or accessory hemiazygos veins, as examples. In still other examples, a transvenous lead may be advanced to position electrodes within the heart, e.g., within an atrial and/or ventricular heart chamber.

(25) More generally, the disclosed techniques may be used in conjunction with any device that is configured to determine a rate or intervals of sensed cardiac events, which may include implantable or external pacemakers and defibrillators and implantable or external heart rate monitors, which may use skin or surface electrodes for sensing cardiac electrical signals. The techniques disclosed herein are not dependent on the particular type of sensing electrodes used or their position, either internal or external. The medical devices shown in FIGS. 1A-3 are examples of medical devices that may be implemented in a system performing techniques disclosed herein with no limitation intended.

(26) FIGS. 1A and 1B are conceptual diagrams of a medical device system **10** configured to sense cardiac events from a cardiac electrical signal and deliver cardiac electrical stimulation therapies according to one example. System **10** includes an ICD **14** connected to a non-transvenous, extra-cardiovascular electrical stimulation and sensing lead **16**. FIG. 1A is a front view of ICD **14** implanted within patient **12**. FIG. 1B is a side view of ICD **14** implanted within patient **12**. FIGS. 1A and 1B are described in the context of an ICD system **10** capable of providing high voltage CV/DF shocks and in some examples cardiac pacing pulses.

(27) ICD **14** includes a housing **15** that forms a hermetic seal that protects internal components of ICD **14**. The housing **15** of ICD **14** may be formed of a conductive material, such as titanium or titanium alloy. The housing **15** may function as an electrode (sometimes referred to as a “can” electrode). Housing **15** may be used as an active can electrode for use in delivering CV/DF shocks or other high voltage pulses delivered using a high voltage therapy circuit. In other examples, housing **15** may be available for use in delivering unipolar, low voltage cardiac pacing pulses and/or for sensing cardiac electrical signals in combination with electrodes carried by lead **16**. In other instances, the housing **15** of ICD **14** may include a plurality of electrodes on an outer portion

of the housing. The outer portion(s) of the housing **15** functioning as an electrode(s) may be coated with a material, such as titanium nitride, e.g., for reducing post-stimulation polarization artifact. (28) ICD **14** includes a connector assembly **17** (also referred to as a connector block or header) that includes electrical feedthroughs crossing housing **15** to provide electrical connections between conductors extending within the lead body **18** of lead **16** and electronic components included within the housing **15** of ICD **14**. As will be described in further detail herein, housing **15** may house one or more processors, memories, transceivers, cardiac electrical signal sensing circuitry, therapy delivery circuitry, power sources and other components for sensing cardiac electrical signals, detecting a heart rhythm, and controlling and delivering electrical stimulation pulses to treat an abnormal heart rhythm.

(29) Lead **16** is shown in this example as an extracardiovascular lead implanted outside the ribcage and sternum. Lead **16** includes an elongated lead body **18** having a proximal end **27** that includes a lead connector (not shown) configured to be connected to ICD connector assembly **17** and a distal portion **25** that includes one or more electrodes. In the example illustrated in FIGS. **1A** and **1B**, the distal portion **25** of lead body **18** includes defibrillation electrodes **24** and **26** and pace/sense electrodes **28** and **30**. In some cases, defibrillation electrodes **24** and **26** may together form a defibrillation electrode in that they may be configured to be activated concurrently. Alternatively, defibrillation electrodes **24** and **26** may form separate defibrillation electrodes in which case each of the electrodes **24** and **26** may be activated independently.

(30) Electrodes **24** and **26** (and in some examples housing **15**) are referred to herein as defibrillation electrodes because they may be utilized, individually or collectively, for delivering high voltage stimulation therapy (e.g., cardioversion or defibrillation shocks). Electrodes **24** and **26** may be elongated coil electrodes and generally have a relatively high surface area for delivering high voltage electrical stimulation pulses compared to pacing and sensing electrodes **28** and **30**. However, electrodes **24** and **26** and housing **15** may also be utilized to provide pacing functionality, sensing functionality or both pacing and sensing functionality in addition to or instead of high voltage stimulation therapy. In this sense, the use of the term “defibrillation electrode” herein should not be considered as limiting the electrodes **24** and **26** for use in only high voltage cardioversion/defibrillation shock therapy applications. For example, either of electrodes **24** and **26** may be used as a sensing electrode in a sensing electrode vector for sensing cardiac electrical signals and determining a need for an electrical stimulation therapy.

(31) Electrodes **28** and **30** are relatively smaller surface area electrodes which are available for use in sensing electrode vectors for sensing cardiac electrical signals and may be used for delivering relatively low voltage cardiac pacing pulses in some configurations. Electrodes **28** and **30** are referred to herein as “pace/sense electrodes” because they are generally configured for use in low voltage applications, e.g., used as either a cathode or anode for delivery of pacing pulses and/or sensing of cardiac electrical signals, as opposed to delivering high voltage CV/DF shocks. In some instances, electrodes **28** and **30** may provide only pacing functionality, only sensing functionality or both.

(32) ICD **14** may sense cardiac electrical signals corresponding to electrical activity of heart **8** via one or more sensing electrode vectors that include combinations of electrodes **24**, **26**, **28** and/or **30**. In some examples, housing **15** of ICD **14** is used in combination with one or more of electrodes **24**, **26**, **28** and/or **30** in a sensing electrode vector. In the example illustrated in FIGS. **1A** and **1B**, electrode **28** is located proximal to defibrillation electrode **24**, and electrode **30** is located between defibrillation electrodes **24** and **26**. One, two or more pace/sense electrodes may be carried by lead body **18**. For instance, a third pace/sense electrode may be located distal to defibrillation electrode **26** in some examples. Electrodes **28** and **30** are illustrated as ring electrodes; however, electrodes **28** and **30** may comprise any of a number of different types of electrodes, including ring electrodes, short coil electrodes, hemispherical electrodes, directional electrodes, segmented electrodes, or the like. Electrodes **28** and **30** may be positioned at other locations along lead body **18** and are not

limited to the positions shown. In other examples, lead **16** may include fewer or more pace/sense electrodes and/or defibrillation electrodes than the example shown here.

(33) Lead **16** may extend subcutaneously or submuscularly over the ribcage **32** medially from the connector assembly **27** of ICD **14** toward a center of the torso of patient **12**, e.g., toward xiphoid process **20** of patient **12**. At a location near xiphoid process **20**, lead **16** may bend or turn to extend superiorly, subcutaneously or submuscularly, over the ribcage and/or sternum, substantially parallel to sternum **22**. Although illustrated in FIG. **1A** as being offset laterally from and extending substantially parallel to sternum **22**, the distal portion **25** of lead **16** may be implanted at other locations, such as over sternum **22**, offset to the right or left of sternum **22**, angled laterally from sternum **22** toward the left or the right, or the like. Alternatively, lead **16** may be placed along other subcutaneous or submuscular paths. The path of non-transvenous, extra-cardiovascular lead **16** may depend on the location of ICD **14**, the arrangement and position of electrodes carried by the lead body **18**, and/or other factors. The techniques disclosed herein are not limited to a particular path of lead **16** or final locations of electrodes **24**, **26**, **28** and **30**.

(34) Electrical conductors (not illustrated) extend through one or more lumens of the elongated lead body **18** of lead **16** from the lead connector at the proximal lead end **27** to electrodes **24**, **26**, **28**, and **30** located along the distal portion **25** of the lead body **18**. The elongated electrical conductors contained within the lead body **18**, which may be separate respective insulated conductors within the lead body **18**, are each electrically coupled with respective defibrillation electrodes **24** and **26** and pace/sense electrodes **28** and **30**. The respective conductors electrically couple the electrodes **24**, **26**, **28**, and **30** to circuitry, such as a therapy delivery circuit and/or a cardiac electrical signal sensing circuit, of ICD **14** via connections in the connector assembly **17**, including associated electrical feedthroughs crossing housing **15**. The electrical conductors transmit therapy from a therapy delivery circuit within ICD **14** to one or more of defibrillation electrodes **24** and **26** and/or pace/sense electrodes **28** and **30** and transmit cardiac electrical signals sensed from the patient's heart **8** from one or more of defibrillation electrodes **24** and **26** and/or pace/sense electrodes **28** and **30** to the sensing circuit within ICD **14**.

(35) The lead body **18** of lead **16** may be formed from a non-conductive material, including silicone, polyurethane, fluoropolymers, mixtures thereof, and/or other appropriate materials, and shaped to form one or more lumens within which the one or more conductors extend. Lead body **18** may be tubular or cylindrical in shape. In other examples, the distal portion **25** (or all of) the elongated lead body **18** may have a flat, ribbon or paddle shape. Lead body **18** may be formed having a preformed distal portion **25** that is generally straight, curving, bending, serpentine, undulating or zig-zagging.

(36) In the example shown, lead body **18** includes a curving distal portion **25** having two “C” shaped curves, which together may resemble the Greek letter epsilon, “ ϵ .” Defibrillation electrodes **24** and **26** are each carried by one of the two respective C-shaped portions of the lead body distal portion **25**. The two C-shaped curves are seen to extend or curve in the same direction away from a central axis of lead body **18**, along which pace/sense electrodes **28** and **30** are positioned. Pace/sense electrodes **28** and **30** may, in some instances, be approximately aligned with the central axis of the straight, proximal portion of lead body **18** such that mid-points of defibrillation electrodes **24** and **26** are laterally offset from pace/sense electrodes **28** and **30**.

(37) Other examples of extra-cardiovascular leads including one or more defibrillation electrodes and one or more pacing and sensing electrodes may include a curving, serpentine, undulating or zig-zagging distal portion of the lead body **18**. The techniques disclosed herein are not limited to any particular lead body design, however. In other examples, lead body **18** is a flexible elongated lead body without any pre-formed shape, bends or curves.

(38) ICD **14** analyzes the cardiac electrical signals received from one or more sensing electrode vectors to monitor for abnormal rhythms, such as bradycardia, ventricular tachycardia (VT) or ventricular fibrillation (VF). ICD **14** may analyze the rate of sensed cardiac events and the

morphology of the cardiac electrical signals to monitor for tachyarrhythmia in accordance with any of a number of tachyarrhythmia detection techniques. ICD **14** generates and delivers electrical stimulation therapy in response to detecting a tachyarrhythmia (e.g., VT or VF) using a therapy delivery electrode vector which may be selected from any of the available electrodes **24**, **26**, **28** **30** and/or housing **15**. ICD **14** may deliver anti-tachycardia pacing (ATP) in response to VT detection and in some cases may deliver ATP prior to a CV/DF shock or during high voltage capacitor charging in an attempt to avert the need for delivering a CV/DF shock. If ATP does not successfully terminate VT or when VF is detected, ICD **14** may deliver one or more CV/DF shocks via one or both of defibrillation electrodes **24** and **26** and/or housing **15**. ICD **14** may deliver the CV/DF shocks using electrodes **24** and **26** individually or together as a cathode (or anode) and with the housing **15** as an anode (or cathode). ICD **14** may generate and deliver other types of electrical stimulation pulses such as post-shock pacing pulses, asystole pacing pulses, or bradycardia pacing pulses using a pacing electrode vector that includes one or more of the electrodes **24**, **26**, **28**, and **30** and the housing **15** of ICD **14**.

(39) ICD **14** is shown implanted subcutaneously on the left side of patient **12** along the ribcage **32**. ICD **14** may, in some instances, be implanted between the left posterior axillary line and the left anterior axillary line of patient **12**. ICD **14** may, however, be implanted at other subcutaneous or submuscular locations in patient **12**. For example, ICD **14** may be implanted in a subcutaneous pocket in the pectoral region. In this case, lead **16** may extend subcutaneously or submuscularly from ICD **14** toward the manubrium of sternum **22** and bend or turn and extend inferiorly from the manubrium to the desired location subcutaneously or submuscularly. In yet another example, ICD **14** may be placed abdominally. Lead **16** may be implanted in other extra-cardiovascular locations as well. For instance, as described with respect to FIGS. **2A-2C**, the distal portion **25** of lead **16** may be implanted underneath the sternum/ribcage in the substernal space.

(40) An external device **40** is shown in telemetric communication with ICD **14** by a communication link **42**. External device **40** may include a processor **52**, memory **53**, display unit **54**, user interface **56** and telemetry unit **58**. Processor **52** controls external device operations and processes data and signals received from ICD **14**.

(41) Processor **52** executes instructions stored in memory **53**. Processor **52** may include any one or more of a microprocessor, a controller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or equivalent discrete or analog logic circuitry. In some examples, processor **52** may include multiple components, such as any combination of one or more microprocessors, one or more controllers, one or more DSPs, one or more ASICs, or one or more FPGAs, as well as other discrete or integrated logic circuitry. The functions attributed to processor **52** herein may be embodied as software, firmware, hardware or any combination thereof.

(42) Display **54**, which may include a graphical user interface (GUI), displays data and other information to a user for reviewing ICD operation and programmed parameters as well as cardiac electrical signals retrieved from ICD **14**. As described below, processor **52** may receive sensed cardiac electrical signals from ICD **14** for processing and analysis according to the techniques disclosed herein. External device processor **52** may be configured to determine sensed cardiac events from a cardiac electrical signal corresponding to a specified sensing electrode vector and according to one or more different sensing control parameters used to set the cardiac event sensing threshold amplitude. Processor **52** may be configured to determine a cardiac event rate or cardiac event intervals based on the determined sensed cardiac events. Based on the determined cardiac event rate or cardiac event intervals, processor **52** may determine an acceptable setting of a sensing control parameter for reliable cardiac event sensing.

(43) Based on the determined cardiac event intervals, for example, processor **52** may determine whether a tachyarrhythmia detection is expected to be made from the determined cardiac events and may determine a time interval until a predicted tachyarrhythmia detection by ICD **14**.

Processor **52** may generate a display of data related to the processing and analysis of cardiac electrical signals on display unit **54**, as further described below. The generated data display may include an acceptable or recommended setting for one or more sensing control parameters determined by processor **52**. The generated display may be a GUI that a user may interact with for selecting sensing control parameters to be applied for determining sensed cardiac events and/or for authorizing and initiating programming of recommended settings determined by processor **52**.

(44) Memory **53** may include any volatile, non-volatile, magnetic, optical, or electrical media, such as a random access memory (RAM), read-only memory (ROM), non-volatile RAM (NVRAM), electrically-erasable programmable ROM (EEPROM), flash memory, or any other digital or analog media. Memory **53** may be configured to store sensing control parameters and associated programmable settings. Memory **53** may store sensed cardiac event data determined by processor **52** for use in generating an output representative of the determined sensed cardiac events as disclosed herein.

(45) User interface **56** may include a mouse, touch screen, keypad or the like to enable a user to interact with external device **40** to initiate a telemetry session with ICD **14** for retrieving data from and/or transmitting data to ICD **14**, including programmable parameters for controlling cardiac event sensing and therapy delivery. A clinician may use user interface **56** to send and receive commands to ICD **14** via external device **40**. As described herein, a clinician may use user interface **56** to specify one or more cardiac event sensing control parameters. Typically, user interface **56** includes one or more input devices and one or more output devices, including display unit **54**. The input devices of user interface **56** may include a communication device such as a network interface, keyboard, pointing device, voice responsive system, video camera, biometric detection/response system, button, sensor, mobile device, control pad, microphone, presence-sensitive screen, touch-sensitive screen (which may be included in display unit **54**), network, or any other type of device for detecting input from a human or machine.

(46) The one or more output devices of user interface **56** may include a communication unit such as a network interface, display, sound card, video graphics adapter card, speaker, presence-sensitive screen, one or more USB interfaces, video and/or audio output interfaces, or any other type of device capable of generating tactile, audio, video, or other output. Display unit **54** may function as an input and/or output device using technologies including liquid crystal displays (LCD), quantum dot display, dot matrix displays, light emitting diode (LED) displays, organic light-emitting diode (OLED) displays, cathode ray tube displays, e-ink, or monochrome, color, or any other type of display capable of generating tactile, audio, and/or visual output. In other examples, user interface **56** may produce an output to a user in another fashion, such as via a sound card, video graphics adapter card, speaker, presence-sensitive screen, touch-sensitive screen, one or more USB interfaces, video and/or audio output interfaces, or any other type of device capable of generating tactile, audio, video, or other output. In some examples, display unit **54** is a presence-sensitive display that may serve as a user interface device that operates both as one or more input devices and one or more output devices.

(47) Telemetry unit **58** includes a transceiver and antenna configured for bidirectional communication with a telemetry circuit included in ICD **14** and is configured to operate in conjunction with processor **52** for sending and receiving data relating to ICD functions via communication link **42**. Communication link **42** may be established between ICD **14** and external device **40** using a radio frequency (RF) link such as BLUETOOTH®, Wi-Fi, or Medical Implant Communication Service (MICS) or other RF or communication frequency bandwidth or communication protocols. Data stored or acquired by ICD **14**, including cardiac electrical signals or associated data derived therefrom, results of device diagnostics, and histories of detected rhythm episodes and delivered therapies, may be retrieved from ICD **14** by external device **40** following an interrogation command. In particular, external device **40** may retrieve episodes of sensed cardiac electrical signals from ICD **14**.

(48) For example, as described below, external device **40** may retrieve an episode of at least two different cardiac electrical signals sensed by ICD **14** using at least two different sensing electrode vectors for processing and analysis by processor **52**. Processor **52** may construct at least one cardiac electrical signal from two or more sensed cardiac electrical signals received from ICD **14**. Processor **52** may determine a sensed cardiac event rate and/or sensed cardiac event intervals from each cardiac electrical signal, sensed and constructed, based on one or more different cardiac event sensing threshold control parameter settings for use in determining a recommended sensing control parameter setting. The processing and analysis performed by processor **52** may be performed in real time as cardiac electrical signals are sensed by ICD **14** and transmitted to external device **40**. Alternatively, processor **40** may perform processing and analysis of previously sensed cardiac electrical signal episodes that are stored by ICD **14** and transmitted to external device **40**.

(49) External device **40** may be used to program sensing control parameters, cardiac rhythm detection parameters and therapy delivery control parameters used by ICD **14**. External device **40** may be embodied as a programmer used in a hospital, clinic or physician's office to retrieve data from ICD **14** and to program operating parameters and algorithms in ICD **14** for controlling ICD functions. External device **40** may alternatively be embodied as a home monitor or handheld device, which may be a tablet, cell phone or other personal device. While external device **40** is shown only in FIG. **1A**, it is to be understood that portions of the techniques disclosed herein may be performed by an external device, such as device **40**, configured to communicate with an implantable or another external medical device configured to sense and transmit cardiac electrical signals to external device **40**. Aspects of external device **40** may generally correspond to the external programming/monitoring unit disclosed in U.S. Pat. No. 5,507,782 (Kieval, et al.), hereby incorporated herein by reference in its entirety. An example programmer that may be configured to perform the techniques disclosed herein is the CARELINK® Programmer, commercially available from Medtronic, Inc., Minneapolis, Minnesota, USA.

(50) In some examples, external device **40** includes external ports **55** adapted to an interface of one or more external leads that include electrodes for sensing an electrocardiogram. In some examples, external device **40** may receive, via the one or more electrodes of the one or more external leads, electrocardiogram data from the patient or from another electrocardiogram input source. The electrocardiogram may be received by processor **52** for determining a heart rate, which may be used in establishing expected cardiac rate criteria for comparing to determined sensed cardiac event data in identifying acceptable sensing control parameters. In some examples, lead **16** may be coupled to external device **40**, e.g., via external ports **55** and any necessary electrical connectors or adaptors. In this case, external device **40** is configured to receive cardiac electrical signals sensed via electrodes carried by lead **16** prior to coupling lead **16** to ICD **14**, without requiring wireless transmission of the sensed cardiac electrical signals to external device **40**. A subcutaneous or cutaneous patch electrode or other electrode may be temporarily positioned at an approximate implant site of ICD **14** to serve as a substitute electrode for ICD housing **15**. At least two sensing electrode vectors may be selected from the electrodes carried by lead **16**. In some examples, the electrode positioned temporarily as a substitute for the ICD housing **15** may be selected in one or both of the sensing electrode vectors to approximate a sensing electrode vector that includes ICD housing **15** and an electrode carried by lead **16**. Processing and analysis of cardiac electrical signals sensed using lead **16** for selecting sensing control parameters may therefore be performed by external device processor **52** prior to connecting and implanting ICD **14**. In this way, a clinician can verify that an acceptable implant position of lead **16** associated with acceptable settings of the sensing control parameters can be achieved for reliably sensing cardiac event signals and tachyarrhythmias based on output generated by external device processor **52**, prior to implanting ICD **14**.

(51) FIG. **1C** is a conceptual diagram of medical device system **10** and an external cardiac monitor **60**. Cardiac monitor **60** may be coupled to multiple surface electrodes **31** for sensing surface

electrocardiogram (ECG) signals. In some examples, the surface electrodes **31** may be positioned over and in approximate alignment with implanted electrodes coupled to ICD **14** or approximate locations of electrodes that are to be implanted and coupled to an ICD. The locations of surface electrodes **31** as shown in FIG. **1C** are illustrative in nature. Surface electrodes **31** may be positioned at other locations corresponding to approximate subcutaneous, substernal or intravenous locations of implanted electrodes or targeted locations of implantable electrodes that are not yet implanted. Surface ECG signals may be acquired by cardiac monitor **60**, e.g., during an induced tachyarrhythmia during an office visit or implant procedure or during a spontaneous tachyarrhythmia during ambulatory monitoring of patient **12**.

(52) A processor included in cardiac monitor **60** may receive, process and analyze the surface ECG signals to determine acceptable or recommended sensing control parameter settings for use by the ICD **14** when coupled to implantable electrodes corresponding to locations and associated sensing electrode vectors simulated by the surface electrode locations. In other examples, the surface ECG signals may be transmitted to external device **40** for processing and analysis by external device processor **52**. The surface ECG signals may be used by the processor of cardiac monitor **60** or external device **40** to construct at least one additional ECG signal corresponding to a different sensing electrode vector than the vectors used to sense the surface ECG signals. The constructed ECG signal may be correlated to a sensing electrode vector between implantable electrodes that are or will be coupled to ICD **14**. The constructed ECG signal be analyzed according to the techniques disclosed herein for determining recommended or acceptable sensing control parameters that may be programmed into ICD **14** for use in sensing cardiac signals and detecting tachyarrhythmia. In this manner, acceptable sensitivity control parameter settings could be identified prior to implanting ICD **14** and/or lead **16**. This technique may be utilized as part of a pre-screening procedure alone or in conjunction with other screening criteria to ensure a patient is an appropriate candidate for the system.

(53) In addition to or alternatively to sensed surface ECG signals, cardiac monitor **60** may be capable of wireless communication with ICD **14** for receiving cardiac electrical signals sensed by ICD **14**. For example, cardiac monitor **60** may be coupled to an inductive antenna **33** that may be positioned over ICD **14**, which may be implanted beneath the skin, to receive cardiac electrical signals sensed by ICD **14**. The cardiac electrical signals may be received by cardiac monitor **60** from ICD **14** in real time via antenna **33**. Cardiac monitor **60** may communicate with ICD **14** using other communication protocols, such as RF communication, whether that be proprietary or utilizing non-proprietary protocol (e.g., Bluetooth).

(54) As described below in conjunction with FIGS. **4** and **5**, ICD **14** includes cardiac electrical signal sensing circuitry which may be configured to sense multiple cardiac electrical signals simultaneously, each using a different sensing electrode vector selected from different combinations of the available electrodes **24**, **26**, **28**, **30** and housing **15**. However, due to processing and memory limitations of an implantable device, ICD **14** may have limited capacity for storing multiple cardiac electrical signals and/or limited capacity for storing signals sampled at relatively high sampling rates over several seconds or minutes. For example, ICD **14** may be configured to store up to 10, 20, 30, 40, 60, 100 or 120 seconds of at least two wideband filtered cardiac electrical signals sampled at a frequency of 128 Hz in its internal memory. These stored signals may be analyzed by ICD **14** or transmitted to external device **40** for post-processing and analysis for determining recommended or acceptable sensing control parameters according to the techniques disclosed herein. However, ICD **14** may be capable of sensing and transmitting more than two cardiac electrical signals in real time, e.g., three, four (or more) cardiac electrical signals sensed using three, four (or more) different sensing electrode vectors, and/or sensing and transmitting cardiac electrical signals at a relatively higher sampling rate, e.g., 256 Hz. Cardiac electrical signals sensed from more sensing electrode vectors and/or at relatively higher sampling rates than the storage capacity of ICD internal memory may be transmitted to an external device, e.g., external

cardiac monitor **60** or external device **40**, to enable analysis of a greater number of sensing electrode vectors.

(55) The sensed cardiac electrical signals may be transmitted from ICD **14** to external device **40** in real time, for example during an office visit or a tachyarrhythmia induction procedure, to enable post-processing and analysis of multiple signals by processor **52** of external device **40**. This analysis of real-time transmitted signals may be limited to cardiac signal episodes obtained during the office visit, e.g., during a tachyarrhythmia induction. In order to enable processing and analysis of cardiac electrical signals acquired during a spontaneous tachyarrhythmia episode detected by ICD **14**, without the limitations posed by the storage capacity of ICD internal memory, ICD **14** may be configured to transmit real-time sensed cardiac electrical signals to external cardiac monitor **60**, e.g., during ambulatory monitoring. External cardiac monitor **60** may be configured to receive cardiac electrical signals transmitted from ICD **14** via antenna **33**, store the cardiac electrical signal data, e.g., in episodes of several seconds or minutes in duration, and transmit the data to external device **40** for post-processing and analysis according to the techniques disclosed herein. In this way, an external processor which may be included in cardiac monitor **60** or external device processor **52**, is enabled to analyze a greater number of sensed cardiac electrical signals, determine a greater number of constructed cardiac electrical signals from the sensed signals, and/or process and analyze sensed cardiac signals acquired at a relatively higher sampling rate.

(56) For example, instead of storing two cardiac electrical signals at a sampling rate of 128 Hz in ICD **14**, which enables one additional cardiac electrical signal to be constructed from the two sensed signals, external cardiac monitor **60** may receive four cardiac electrical signals from ICD **14**, sensed using four different sensing electrode vectors at a higher sampling rate of 256 Hz. The four sensed signals may be transmitted in real time to external cardiac monitor **60**, e.g., during an implant procedure, during an office visit, during a tachyarrhythmia induction procedure or when ICD **14** is detecting a spontaneous ventricular tachyarrhythmia, allowing sensed cardiac signal data to be acquired and stored beyond the limits of ICD **14** memory capacity.

(57) Processor **52** of external device **40** may receive episodes of the four cardiac electrical signals, which may correspond to induced or spontaneous tachyarrhythmia episodes, from external cardiac monitor **60** via communication link **35**, which may be a wired or wireless communication link, e.g., using RF telemetry. Processor **52** of external device **40** may be configured to construct up to four additional cardiac electrical signals, corresponding to four sensing electrode vectors different from the four sensing electrode vectors used to sense the four sensed cardiac electrical signals such that all eight possible sensing electrode vectors selectable from electrodes **24**, **26**, **28**, **30** and housing **15** may be analyzed for determining recommended sensing control parameters according to the techniques disclosed herein. As such, in some examples, external device processor **52** may receive the sensed cardiac electrical signals from ICD **14** via cardiac monitoring device **60**. In other examples, a processor included in cardiac monitoring device **60** may receive the cardiac electrical signals sensed by ICD **14** via antenna **33** for processing and analysis by cardiac monitoring device **60**. Recommended or acceptable sensing control parameters determined by the cardiac monitoring device **60** or external device **40** may be displayed to a user, e.g., in a user interface as described below in conjunction with FIGS. **11**, **12** and **14-16**.

(58) Cardiac monitoring device **60** may be a wearable or portable monitoring device, such as a Holter monitor or mobile cardiac telemetry unit. Cardiac monitoring device **60** may optionally be coupled to surface or skin electrodes **31** for acquiring and storing or transmitting ECG signals to a remote patient monitoring database. It is to be understood, however, that ECG signal sensing by cardiac monitoring device **60** is optional in some examples. Cardiac monitoring device **60** may be configured to receive sensed cardiac electrical signals from ICD **14** without necessarily acquiring ECG signals from surface electrodes **31**. As such cardiac monitoring device **60** may function as extended memory capacity of medical device system **10** for storing cardiac electrical signal data received from ICD **14** and as a relay device for transmitting the stored data received from ICD **14**

to external device **40** or another computer for processing the received signals.

(59) Cardiac monitoring device **60** may be configured to receive episodes of at least two cardiac electrical signals sensed by ICD **14** during one or more detected tachyarrhythmia episodes. The received cardiac signal episodes may be stored in memory of cardiac monitoring device **60** for later transmission to external device **40** via communication link **35**, transmitted in real time to a remote patient monitoring computer or database for subsequent analysis or processed and analyzed by a processor included in cardiac monitoring device **60**. Cardiac monitoring device **60** may be configured to receive cardiac electrical signals from ICD **14** that are sensed during multiple, different tachyarrhythmia episodes detected by ICD **14**. In this way, the techniques disclosed herein for determining acceptable or recommended sensing control parameters for tachyarrhythmia detection by ICD **14** may be applied to a greater number of sensed signals and/or signals acquired at a higher sampling rate during one or more spontaneous tachyarrhythmia episodes.

(60) FIGS. 2A-2C are conceptual diagrams of patient **12** implanted with ICD system **10** in a different implant configuration than the arrangement shown in FIGS. 1A-1B. FIG. 2A is a front view of patient **12** implanted with ICD system **10**. FIG. 2B is a side view of patient **12** implanted with ICD system **10**. FIG. 2C is a transverse view of patient **12** implanted with ICD system **10**. In this arrangement, lead **16** of system **10** is implanted at least partially underneath sternum **22** of patient **12**. Lead **16** extends subcutaneously or submuscularly from ICD **14** toward xiphoid process **20** and at a location near xiphoid process **20** bends or turns and extends superiorly within anterior mediastinum **36** in a substernal position.

(61) Anterior mediastinum **36** may be viewed as being bounded laterally by pleurae **39**, posteriorly by pericardium **38**, and anteriorly by sternum **22** (see FIG. 2C). The distal portion **25** of lead **16** may extend along the posterior side of sternum **22** substantially within the loose connective tissue and/or substernal musculature of anterior mediastinum **36**. A lead implanted such that the distal portion **25** is substantially within anterior mediastinum **36**, or within a pleural cavity or more generally within the thoracic cavity, may be referred to as a “substernal lead.”

(62) In the example illustrated in FIGS. 2A-2C, lead **16** is located substantially centered under sternum **22**. In other instances, however, lead **16** may be implanted such that it is offset laterally from the center of sternum **22**. In some instances, lead **16** may extend laterally such that distal portion **25** of lead **16** is underneath/below the ribcage **32** in addition to or instead of sternum **22**. In other examples, the distal portion **25** of lead **16** may be implanted in other extra-cardiovascular, intra-thoracic locations, including the pleural cavity or around the perimeter of and adjacent or within the pericardium **38** of heart **8**. In the examples described herein in conjunction with FIGS. 1A-2C, electrodes for sensing cardiac electrical signals are carried by a lead that may be advanced to a supra-diaphragmatic position, which may be within the thoracic cavity or outside the thorax in various examples.

(63) FIG. 3 is a conceptual diagram of a medical device system **100** that may be configured to perform techniques disclosed herein according to another example. System **100** includes ICD **114** coupled to transvenous leads **116** and **118** in communication with the right atrium (RA) and right ventricle (RV) of heart **8**. ICD **114** includes a housing **115** enclosing circuitry, such as a processor, telemetry circuitry, sensing circuitry and therapy delivery circuitry as described below in conjunction with FIG. 4. ICD **114** includes connector assembly **117** having connector bores for receiving proximal connectors of RA lead **116** and RV lead **118** and providing electrical connection between electrodes carried by leads **116** and **118** and internal ICD circuitry.

(64) RA lead **116** may carry a distal tip electrode **120** and ring electrode **122** for sensing atrial electrical signals and producing an atrial intra-cardiac electrogram (EGM) signal. RA electrodes **120** and **122** may be used for delivering RA pacing pulses. RV lead **118** may carry pacing and sensing electrodes **132** and **134** for sensing a ventricular electrical signal and producing an RV EGM signal. RV electrodes **132** and **134** may be used to deliver RV pacing pulses. RV lead **118** may also carry an RV defibrillation electrode **124** and a superior vena cava (SVC) defibrillation

electrode **126**. Defibrillation electrodes **124** and **126** are shown as coil electrodes spaced apart proximally from the distal pacing and sensing electrodes **132** and **134**. While RA lead **116** and RV lead **118** are both shown advanced within a respective heart chamber, in some examples, a transvenous lead may be advanced to position electrodes within a venous location, outside the heart.

(65) ICD **114** may be configured to provide dual chamber sensing and pacing therapies as well as high voltage CV/DF shock therapies in response to detecting VT or VF. In other examples, ICD **114** may be configured to provide multi-chamber sensing and pacing therapies, including cardiac resynchronization therapy (CRT), in which case a coronary sinus lead may be advanced along a cardiac vein to position electrodes for sensing and pacing the left ventricle of heart **8**. In still other examples, ICD **114** may be a single chamber device coupled to a single lead, e.g., lead **116** or lead **118**, for sensing cardiac electrical signals and delivery electrical stimulation therapies. ICD **114** may be configured to sense at least two cardiac electrical signals that may be processed and analyzed according to the techniques disclosed herein for determining acceptable sensing control parameters. A processor included in ICD **114** may perform the processing and analysis, or ICD **114** may transmit episodes of the two cardiac electrical signals to another device, e.g., external device **40** shown in FIG. **1A**, for processing and analysis for determining recommended or acceptable sensing control parameters.

(66) FIG. **4** is a conceptual diagram of a medical device configured to sense cardiac electrical signals according to one example. FIG. **4** is described in conjunction with the ICD **14** of FIGS. **1A-2C**, including therapy delivery capabilities. It is to be understood, however, the circuitry and functionality attributed to circuitry described in conjunction with FIG. **4** may be included, in whole or in part, in any of the example medical devices described or listed herein, such as the ICD **114** shown in FIG. **3**. The ICD housing **15** is shown schematically as an electrode in FIG. **4** because the housing of the medical device may be used as an electrode in a sensing electrode vector for cardiac signal sensing and/or for therapy delivery in some examples. The electronic circuitry enclosed within housing **15** includes software, firmware and hardware that cooperatively monitor cardiac signals, determine when an electrical stimulation therapy is necessary, and deliver therapy as needed according to programmed therapy delivery algorithms and control parameters.

(67) ICD **14** as shown in FIG. **4** includes a control circuit **80**, memory **82**, therapy delivery circuit **84**, cardiac electrical signal sensing circuit **86**, and telemetry circuit **88**. Control circuit **80** communicates, e.g., via a data bus, with therapy delivery circuit **84** and sensing circuit **86** for sensing cardiac event signals, detecting cardiac rhythms, and controlling delivery of cardiac electrical stimulation therapies in response to sensed cardiac event signals. Therapy delivery circuit **84** and sensing circuit **86** are electrically coupled to electrodes **24**, **26**, **28**, **30** and the housing **15**, which may function as a common or ground electrode or as an active can electrode for delivering CV/DF shock pulses or cardiac pacing pulses. As described above electrodes **24**, **26**, **28** and **30** shown in FIG. **4** may be carried by a non-transvenous lead advanced to position electrodes in an extra-cardiac location (as shown in FIGS. **1A-2C**) or by a transvenous lead for positioning electrodes within a blood vessel or an intracardiac location (e.g., electrodes **120**, **122**, **124**, **126**, **132**, and **134** as shown in FIG. **3**). Furthermore, electrodes coupled to the medical device may include multiple housing-based electrodes not carried by a lead in some examples.

(68) A power source **98** provides power to the circuitry of ICD **14**, including each of the components **80**, **82**, **84**, **86**, and **88** as needed. Power source **98** may include one or more energy storage devices, such as one or more rechargeable or non-rechargeable batteries. The connections between power source **98** and each of the other components **80**, **82**, **84**, **86** and **88** are to be understood from the general block diagram of FIG. **4** but are not shown for the sake of clarity. For example, power source **98** may be coupled to one or more charging circuits included in therapy delivery circuit **84** for charging holding capacitors or other charge storage devices included in therapy delivery circuit **84** that are discharged at appropriate times under the control of control

circuit **80** for producing electrical pulses according to a therapy protocol. In other examples, power source **98** may serve as a voltage or current source to therapy delivery circuit **84** without requiring a charge storage device. Power source **98** is also coupled to components of cardiac electrical signal sensing circuit **86**, such as sense amplifiers, analog-to-digital converters, switching circuitry, etc., telemetry circuit **88** and memory **82** as needed.

(69) The circuits shown in FIG. **4** represent functionality included in ICD **14** or another medical device operating according to the techniques disclosed herein and may include any discrete and/or integrated electronic circuit components that implement analog and/or digital circuits capable of producing the functions attributed to ICD **14** herein. Functionality associated with one or more circuits may be performed by separate hardware, firmware or software components, or integrated within common hardware, firmware or software components. For example, cardiac event sensing and determination of sensed cardiac event intervals may be performed cooperatively by sensing circuit **86** and control circuit **80** and may include operations implemented in a processor **81** or other signal processing circuitry included in control circuit **80** executing instructions stored in memory **82**. Control signals such as blanking and timing intervals associated with setting the cardiac event sensing threshold amplitude may be sent from control circuit **80** to sensing circuit **86** according to programmed sensing control parameter settings.

(70) The various circuits of ICD **14** may include an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that execute one or more software or firmware programs, a combinational logic circuit, state machine, or other suitable components or combinations of components that provide the described functionality. The particular form of software, hardware and/or firmware employed to implement the functionality disclosed herein will be determined primarily by the particular system architecture employed in the medical device system and by the particular detection and therapy delivery methodologies employed by the medical device system. Providing software, hardware, and/or firmware to accomplish the described functionality in the context of any modern medical device, given the disclosure herein, is within the abilities of one of skill in the art.

(71) Memory **82** may include any volatile, non-volatile, magnetic, or electrical non-transitory computer readable storage media, such as random access memory (RAM), read-only memory (ROM), non-volatile RAM (NVRAM), electrically-erasable programmable ROM (EEPROM), flash memory, or any other memory device. Furthermore, memory **82** may include non-transitory computer readable media storing instructions that, when executed by one or more processing circuits, cause control circuit **80** and/or other ICD components to perform various functions attributed to ICD **14** or those ICD components. The non-transitory computer-readable media storing the instructions may include any of the media listed above.

(72) Cardiac electrical signal sensing circuit **86** (also referred to herein as “sensing circuit” **86**) may be selectively coupled to electrodes **28**, **30** and/or housing **15** in order to sense electrical activity of the patient's heart. Sensing circuit **86** may additionally be selectively coupled to defibrillation electrodes **24** and/or **26** for use in a sensing electrode vector together or in combination with one or more of electrodes **28**, **30** and/or housing **15**. Sensing circuit **86** may be enabled to selectively receive cardiac electrical signals from at least two different sensing electrode vectors from the available electrodes **24**, **26**, **28**, **30**, and housing **15** in some examples. Sensing circuit **86** may monitor one or more cardiac electrical signals for sensing cardiac events and/or producing digitized cardiac electrical signals passed to control circuit **80** for processing and analysis and/or for further transmission to external device **40** via telemetry circuit **88**. For example, sensing circuit **86** may include switching circuitry for selecting which of electrodes **24**, **26**, **28**, **30**, and housing **15** are coupled to one or more sensing channels of sensing circuit **86**.

(73) As described below in conjunction with FIG. **5**, sensing circuit **86** may be configured to amplify, filter, rectify and digitize or otherwise process the cardiac electrical signal received from each selected sensing electrode vector to improve the signal quality for sensing cardiac electrical

events, such as R-waves or P-waves. Cardiac event detection circuitry included within sensing circuit **86** may include one or more sense amplifiers, filters, rectifiers, threshold detectors, comparators, analog-to-digital converters (ADCs), timers or other analog or digital components configured to sense cardiac events, e.g., R-waves attendant to ventricular depolarizations and P-waves attendant to atrial depolarizations from a sensed cardiac electrical signal received by sensing circuit **86** via the selected sensing electrode vector.

(74) Sensing circuit **86** may control the amplitude of an auto-adjusting cardiac event sensing threshold over each cardiac cycle, e.g., as described below in conjunction with FIG. **6**. Sensing circuit **86** may sense a cardiac event in response to a cardiac electrical signal crossing the sensing threshold. Sensing circuit **86** may produce a cardiac sensed event signal, e.g., an atrial sensed event signal in response to a P-wave sensing threshold crossing or a ventricular sensed event signal in response to an R-wave sensing threshold crossing. The cardiac sensed event signals are passed to control circuit **80**. As described below, various sensing threshold control parameters may be used by sensing circuit **86** to set and adjust the cardiac event sensing threshold during each cardiac cycle. These sensing threshold control parameters may be stored in memory **82** and passed to sensing circuit **86** from control circuit **80** for use by hardware, firmware and/or software of control circuit **80** and/or sensing circuit **86** in controlling the amplitude of the cardiac event sensing threshold.

(75) Control circuit **80** receives the cardiac sensed event signals from sensing circuit **86** for determining sensed cardiac event intervals, e.g., RR intervals (RRIs) and/or PP intervals (PPIs), by timing circuit **90**. An RRI is the time interval between two consecutively sensed R-waves and may be determined between consecutive ventricular sensed event signals received by control circuit **80** from sensing circuit **86**. A PPI is the time interval between two consecutively sensed P-waves and may be determined between consecutive atrial sensed event signals received by control circuit **80** from sensing circuit **86**. Depending on programmed therapies, timing circuit **90** may trigger therapy delivery circuit **84** to generate and deliver an electrical stimulation pulse in response to a sensed event signal and/or start a pacing escape interval timer in response to a sensed event signal and restart the escape interval timer in response to the next sensed event signal. The value of the escape interval timer at the time of the next sensed event signal may be buffered in memory **82** as the sensed cardiac event interval for the associated sensed event signal. In this way, memory **82** may store a series of sensed cardiac event intervals for determining a sensed cardiac event rate.

(76) Timing circuit **90** may include various timers and/or counters used to control the timing of therapy delivery by therapy delivery circuit **84**. In response to expiration of an escape interval timer without receiving a cardiac sensed event signal, control circuit **80** may control therapy delivery circuit **84** to generate and deliver a pacing pulse. Timing circuit **90** may additionally set time windows such as morphology template windows, morphology analysis windows or perform other timing related functions of ICD **14** including synchronizing CV/DF shocks or other therapies delivered by therapy delivery circuit **84** with sensed cardiac events.

(77) Control circuit **80** may include a tachyarrhythmia detector **92** configured to analyze signals received from sensing circuit **86** for detecting tachyarrhythmia. Tachyarrhythmia detector **92** may detect tachyarrhythmia based on cardiac events sensed by sensing circuit **86** meeting tachyarrhythmia detection criteria, such as a threshold number of sensed cardiac events occurring at cardiac event intervals falling in a tachyarrhythmia interval range. Tachyarrhythmia detector **92** may be implemented in control circuit **80** as hardware, software and/or firmware that processes and analyzes signals received from sensing circuit **86** for detecting tachyarrhythmia, e.g., supraventricular tachycardia (SVT), VT and/or VF. Tachyarrhythmia detector **92** may include comparators and counters for counting cardiac event intervals, e.g., PPIs or RRIs determined by timing circuit **90**, that fall into various rate detection zones for determining an atrial rate and/or a ventricular rate or performing other rate- or interval-based assessment of cardiac sensed event signals for detecting and discriminating tachyarrhythmias.

(78) For example, tachyarrhythmia detector **92** may compare the RRs determined by timing circuit **90** to one or more tachyarrhythmia detection interval zones, such as a tachycardia detection interval zone and a fibrillation detection interval zone. RRs falling into a detection interval zone are counted by a respective VT interval counter or VF interval counter and in some cases in a combined VT/VF interval counter included in tachyarrhythmia detector **92**. The VF detection interval threshold may be set to 300 to 350 milliseconds (ms), as an example. For instance, if the VF detection interval is set to 320 ms, RRs that are less than 320 ms are counted by the VF interval counter. When VT detection is enabled, the VT detection interval may be programmed to be in the range of 350 to 420 ms, or 400 ms as an example. RRs that are less than the VT detection interval but greater than the VF detection interval may be counted by a VT interval counter. In order to detect VT or VF, the respective VT or VF interval counter is required to reach a threshold “number of intervals to detect” or “NID.”

(79) As an example, the NID to detect VT may require that the VT interval counter reaches 18 VT intervals, 24 VT intervals, 32 VT intervals or other selected NID. In some examples, the VT intervals may be required to be consecutive intervals, e.g., 18 out of 18, 24 out of 24, or 32 out of the most recent 32 consecutive RRs. The NID required to detect VF may be programmed to a threshold number of X VF intervals out of Y consecutive RRs. For instance, the NID required to detect VF may be 18 VF intervals out of the most recent 24 consecutive RRs or 30 VF intervals out 40 consecutive RRs, as examples. When a VT or VF interval counter reaches an NID, a ventricular tachyarrhythmia may be detected by tachyarrhythmia detector **92**. The NID may be programmable and range from as low as 12 to as high as 40, with no limitation intended. VT or VF intervals may be detected consecutively or non-consecutively out of a specified number of most recent RRs. In some cases, a combined VT/VF interval counter may count both VT and VF intervals and detect a tachyarrhythmia episode based on the fastest intervals detected when a specified NID is reached.

(80) Tachyarrhythmia detector **92** may be configured to perform other signal analysis for determining if other detection criteria are satisfied before detecting VT or VF, such as R-wave morphology criteria and onset criteria. To support additional cardiac signal analyses, sensing circuit **86** may pass a digitized cardiac electrical signal to control circuit **80**, e.g., an ECG signal when sensed using electrodes outside the heart or an EGM signal when sensed using intracardiac electrodes. The digitized cardiac electrical signal may be passed to control circuit **80** for morphology analysis performed by tachyarrhythmia detector **92** for detecting and discriminating heart rhythms. A cardiac electrical signal from a selected sensing electrode vector may be passed through a filter and amplifier, provided to a multiplexer and thereafter converted to a multi-bit digital signal by an analog-to-digital converter, all included in sensing circuit **86**, for storage in memory **82** and/or for real time transmission via telemetry circuit **88**. Memory **82** may include one or more circulating buffers to temporarily store digital cardiac electrical signal segments (also referred to herein as “episodes”) for analysis performed by control circuit **80** and/or by external device processor **52** after transmission via telemetry circuit **88**. Control circuit **80** may be a microprocessor-based controller that employs digital signal analysis techniques to characterize the digitized signals stored in memory **82** to recognize and classify the patient's heart rhythm employing any of numerous signal processing methodologies for analyzing cardiac electrical signals and cardiac event waveforms, e.g., R-waves.

(81) Therapy delivery circuit **84** includes charging circuitry, one or more charge storage devices such as one or more high voltage capacitors and/or low voltage capacitors, and switching circuitry that controls when the capacitor(s) are discharged across a selected pacing electrode vector or CV/DF shock vector. Charging of capacitors to a programmed pulse amplitude and discharging of the capacitors for a programmed pulse width may be performed by therapy delivery circuit **84** according to control signals received from control circuit **80**. For example, timing circuit **90** may include programmable digital counters set by a microprocessor of the control circuit **80** for

controlling the basic pacing time intervals associated with various pacing modes or ATP sequences delivered by ICD **14**. The microprocessor of control circuit **80** may also set the amplitude, pulse width, polarity or other characteristics of the cardiac pacing pulses, which may be based on programmed values stored in memory **82**.

(82) In response to detecting VT or VF, control circuit **80** may schedule a therapy and control therapy delivery circuit **84** to generate and deliver the therapy, such as ATP and/or CV/DF therapy. Therapy can be generated by initiating charging of high voltage capacitors via a charging circuit, both included in therapy delivery circuit **84**. Charging is controlled by control circuit **80** which monitors the voltage on the high voltage capacitors, which is passed to control circuit **80** via a charging control line. When the voltage reaches a predetermined value set by control circuit **80**, a logic signal is generated on a capacitor full line and passed to therapy delivery circuit **84**, terminating charging. A CV/DF pulse is delivered to the heart under the control of the timing circuit **90** by an output circuit of therapy delivery circuit **84** via a control bus. The output circuit may include an output capacitor through which the charged high voltage capacitor is discharged via switching circuitry, e. g., an H-bridge, which determines the electrodes used for delivering the cardioversion or defibrillation pulse and the pulse wave shape.

(83) In some examples, the high voltage therapy circuit configured to deliver CV/DF shock pulses can be controlled by control circuit **80** to deliver pacing pulses, e.g., for delivering ATP, post shock pacing pulses or ventricular pacing pulses. In other examples, therapy delivery circuit **84** may include a low voltage therapy circuit for generating and delivering pacing pulses for a variety of pacing needs.

(84) It is recognized that the methods disclosed herein for processing and analyzing cardiac electrical signals may be implemented in a medical device system that is used for monitoring cardiac electrical signals by sensing circuit **86** and control circuit **80** without necessarily having therapy delivery capabilities or in a pacemaker that monitors cardiac electrical signals and delivers cardiac pacing therapies by therapy delivery circuit **84**, without high voltage therapy capabilities such as CV/DF shock capabilities.

(85) Control parameters utilized by control circuit **80** for sensing cardiac events and controlling therapy delivery may be programmed into memory **82** via telemetry circuit **88**. Telemetry circuit **88** includes a transceiver and antenna for communicating with external device **40** (shown in FIG. **1A**) using RF communication or other communication protocols as described above. Under the control of control circuit **80**, telemetry circuit **88** may receive downlink telemetry from and send uplink telemetry to external device **40**. Telemetry circuit **88** may transmit sensed cardiac electrical signals (and in some cases sensed cardiac event markers and associated cardiac event intervals) to another medical device, e.g., external device **40**, for processing and analysis according to the techniques disclosed herein. In other examples, control circuit **80** may be configured to perform some or all of the analysis of cardiac electrical signals as disclosed herein and may transmit the data resulting from the analysis to external device **40**.

(86) FIG. **5** is a conceptual diagram of circuitry that may be included in sensing circuit **86** of a medical device, e.g., ICD **14**, according to one example. Sensing circuit **86** may be coupled to all available electrodes that may be used in a sensing electrode vector for sensing cardiac electrical signals. Using the example medical device system **10** of FIGS. **1A-2C**, sensing circuit **86** is shown coupled to pace/sense electrodes **28** and **30**, defibrillation electrodes **24** and **26** and housing **15**, which may be selected in any combination as a sensing electrode vector. Sensing circuit **86** may include switching circuit **61** for controlling which electrodes are selected in a pair as a sensing electrode vector that is coupled to prefilter and amplifier **62**. Two or more sensing electrode vectors may be coupled to prefilter and amplifier **62**. In one example, the two pace/sense electrodes **28** and **30** carried by lead **16** are selected as one sensing electrode vector, and one pace/sense electrode **28** or **30** carried by lead **16** is selected in combination with housing **15** as a second sensing electrode vector. A sensing electrode vector may be selected by switching circuit **61** according to control

signals from control circuit **80**. Switching circuit **61** may include a switch array, switch matrix, multiplexer, or any other type of switching device(s) suitable to selectively couple selected electrodes to the input prefilter and amplifier **62**.

(87) Each electrical signal developed across the selected sensing electrode vectors, e.g., electrodes **28** and **30** in a first sensing electrode vector and electrode **30** and housing **15** in a second sensing electrode vector, is received as a differential input signal by the pre-filter and pre-amplifier **62**. Non-physiological high frequency and DC signals may be filtered by a low pass or bandpass filter included in pre-filter and pre-amplifiers **62**, and high voltage signals may be removed by protection diodes included in pre-filter and pre-amplifiers **62**. Pre-filter and pre-amplifier **62** may amplify the pre-filtered signal by a gain of between 10 and 100, and in one example a gain of 17.5, and may convert the differential signal to a single-ended output signal passed to analog-to-digital converter (ADC) **63**. Pre-filter and amplifier **62** may provide anti-alias filtering and noise reduction prior to digitization.

(88) ADC **63** converts the cardiac electrical signal from an analog signal to a digital bit stream. In one example, ADC **63** may be a sigma-delta converter (SDC), but other types of ADCs may be used. In some examples, the output of ADC **63** may be provided to a decimator (not shown), which functions as digital low-pass filter that increases the resolution and reduces the sampling rate of the cardiac electrical signal. The digital output of ADC **63** may be passed to a digital bandpass filter **64** and on to cardiac event detector **66** for sensing cardiac events. Filter **64** may have a relatively narrow bandpass of approximately 13 Hz to 39 Hz for passing cardiac event signals, such as R-waves, typically occurring in this (or other bandpass) frequency range. The narrowband filtered signal may be passed from filter **64** to rectifier **65** to produce a filtered, rectified signal that is received by a cardiac event detector **66** for sensing cardiac events in response to the narrowband filtered and rectified signal crossing a cardiac event sensing threshold amplitude, for example an R-wave sensing threshold amplitude. In some examples, cardiac event detector **66** may include a P-wave detector for producing atrial sensed event signals in response to a P-wave sensing threshold.

(89) Cardiac event detector **66** may include an auto-adjusting sense amplifier, comparator and/or other detection circuitry that compares the filtered and rectified cardiac electrical signal to a cardiac event sensing threshold amplitude and produces a sensed event signal **68**, which may be a ventricular sensed event signal or an atrial sensed event signal, when the filtered, rectified signal crosses the cardiac event sensing threshold outside of any blanking periods applied by sensing circuit **86**. A cardiac event sensing threshold applied by cardiac event detector **66** may be a multi-level sensing threshold, for example as described below in conjunction with FIG. **6**. Multiple sensing threshold control parameters may be used to adjust the amplitude of the cardiac event sensing threshold starting from the expiration of a post-sense (or post-pace) blanking period until the next cardiac event sensing threshold crossing (or expiration of a pacing interval). The techniques described herein are not limited to a specific behavior of the sensing threshold amplitude and numerous cardiac event sensing threshold control parameters, as described below in conjunction with FIG. **6**, may be defined and used to control the cardiac event sensing threshold amplitude over a cardiac cycle, until a cardiac event is sensed or a pacing escape interval expires resulting in a pacing pulse.

(90) In some examples, event detector **66** includes a peak detector, which may include a sample and hold circuit, for detecting the maximum peak amplitude of the rectified signal following a cardiac sensed event signal produced by cardiac event detector **66**. The maximum peak amplitude may be used by cardiac event detector **66** for setting the starting amplitude of the cardiac event sensing threshold based on the detected maximum peak amplitude. The starting cardiac event sensing threshold amplitude may be set by cardiac event detector **66** upon expiration of any post-sense blanking period applied upon detecting a sensing threshold crossing. The cardiac event sensing threshold amplitude may be set to a starting amplitude that is a percentage of the maximum peak amplitude, e.g., 60%, 70%, 80% or other selected percentage. The starting sensing threshold

amplitude may be adjusted according to the sensing threshold control parameters down to a sensing floor, which may be equal to the programmed sensitivity, or until a sensing threshold crossing, whichever occurs first. The sensitivity is the lowest amplitude that the cardiac event sensing threshold may be adjusted to and is therefore the lowest amplitude of the cardiac electrical signal that may be sensed as a cardiac event. The lower the programmed value of the sensitivity, e.g., 0.03 millivolts or less, the more sensitive sensing circuit **86** is to sensing cardiac events. Sensing circuit **86** is relatively less sensitive to sensing cardiac events when the programmed value of the sensitivity is relatively higher, e.g., 0.9 millivolts or higher.

(91) Sensing circuit **86** may include wideband filter **74** for producing a cardiac EGM (sensed by intracardiac electrodes) or ECG (sensed by extra-cardiac electrodes) signal that is passed to control circuit **80**, e.g., signal **78**, for performing morphological analysis of the cardiac signal waveforms. For example, control circuit **80** may perform morphology analysis of the wideband filtered cardiac electrical signal **78** to detect and distinguish R-waves arising from non-sinus tachycardia or fibrillation waves from normally conducted R-waves. Wideband filter **74** may have a bandpass of approximately 2.5 to 100 Hz. In some examples, filter **74** may include a notch filter to attenuate 60 Hz or 50 Hz line noise. The cardiac electrical signal received by the sensing electrode vector between one of pace/sense electrodes **28** or **30** and housing **15** may be passed from ADC **63** to wideband filter **74** for morphological analysis by control circuit **80** for use in tachyarrhythmia detection. A second cardiac electrical signal received by a different sensing electrode vector, e.g., between pace/sense electrodes **28** and **30**, may be passed to narrowband filter **64** and to wideband filter **74**. Both the first and the second wideband filtered cardiac electrical signals may be analyzed for determining acceptable or recommended sensing control parameters by control circuit **80** or by the processor of another medical device, e.g., external device processor **52** (FIG. **1A**).

(92) Sensing circuit **86** may be configured to select one sensing electrode vector out of multiple available sensing electrode vectors for cardiac event sensing by event detector **66**. The sensing electrode vector may be a programmable sensing control parameter. For the sake of illustration, sensing circuit **86** may select one sensing electrode vector from among three different sensing electrode vectors for sensing cardiac events. The three different sensing electrode vectors may be between pace/sense electrodes **28** and **30**, between pace/sense electrode **28** and housing **15**, and between pace/sense electrode **30** and housing **15** in one example.

(93) Sensing circuit **86** may be configured to select two or more sensing electrode vector signals for passing to wideband filter **74**. In some examples, a single wideband filtered signal is used by control circuit **80** for morphological analysis, e.g., for tachyarrhythmia detection. However, multiple wideband filtered vector signals may be analyzed according to the techniques disclosed herein for determining acceptable or recommended sensing control parameters. For the sake of illustration, sensing circuit **86** may be configured to select up to four sensing electrode vectors for producing four different wideband filtered digital cardiac electrical signals **70**, **72**, **76** and **78** from among ten possible sensing electrode vector signals available from the four electrodes **24**, **26**, **28**, **30** and housing **15**. In the example shown, the ten possible sensing electrode vectors are between pace/sense electrodes **28** and **30**, between pace/sense electrode **28** and each of housing **15**, defibrillation electrode **24** and defibrillation electrode **26**, between pace/sense electrode **30** and each of housing **15**, defibrillation electrode **24** and defibrillation electrode **26**, between defibrillation electrodes **24** and **26**, between defibrillation electrode **24** and housing **15**, and between defibrillation electrode **26** and housing **15**.

(94) All four wideband filtered sensing vectors signals **70**, **72**, **76** and **78** may be analyzed according to techniques disclosed herein for determining acceptable or recommended sensing control parameters. In one example, the wideband filtered signals **70**, **72**, **76** and **78** are sampled at 256 Hz and transmitted by telemetry circuit **88** in real time. The transmitted cardiac electrical signals may be received by another device, e.g., external device **40**, for processing and analysis. In other examples, the wideband filtered signals **70**, **72**, **74**, and/or **76** may be stored in memory **82** for

post-processing and analysis by control circuit **80** or for later transmission via telemetry circuit **88**. Because memory **82** may have limited capacity for storing cardiac electrical signals, fewer signals and/or signals sampled at a lower sampling rate may be stored in memory **82** for later analysis or transmission than the number and/or sampling rate of cardiac electrical signals that may be transmitted to external device **40** in real time for subsequent analysis and processing. For example, two wideband filtered signals sampled at a frequency of 128 Hz may be stored in memory **82** for post-processing and analysis, which may include transmission to another device for performing the post-processing and analysis.

(95) A user interacting with external device **40** may select the sensing electrode vector signals that are passed to wideband filter **74** for storage in memory **82** or real time transmission. In illustrative examples presented herein, at least two sensing electrode vector signals are selected for wideband filtering for real time transmission or storage in memory **82**. The two sensing electrode vectors include one electrode common to both sensing electrode vectors such that a third cardiac electrical signal that would be sensed using a third sensing electrode vector between the two electrodes that are unique to the two selected sensing electrode vectors can be constructed. In this way by sensing two cardiac electrical signals, three cardiac electrical signals corresponding to three different sensing electrode vectors are available for processing and analysis for determining acceptable or recommended sensing control parameters. In the illustrative examples presented herein, the two sensed cardiac electrical signal correspond to a first sensing electrode vector between pace/sense electrodes **28** and **30** and second sensing electrode vector between pace/sense electrode **30** and housing **15**. During processing and analysis, control circuit **80** or another processor, e.g., external device processor **52** (FIG. 1A), may construct a third cardiac electrical signal that is expected to be sensed between the two electrodes that are not common to the sensed cardiac electrical signal sensing vectors, which would be between pace/sense electrode **28** and housing **15** in the illustrative example given here.

(96) By programming selected sensing electrode vector signals for filtering by wideband filter **74**, one or more additional sensing electrode vectors signals may be constructed by a processor, thereby enabling analysis of multiple sensed and constructed cardiac electrical signals for determining acceptable or recommended sensing control parameters, which may include an acceptable or recommended the sensing electrode vector. The techniques disclosed herein for determining acceptable or recommended sensing control parameters, therefore, do not require all cardiac electrical signals that undergo analysis to be sensed cardiac electrical signals. The number of sensing electrode vector signals available for analysis can be increased by constructing one or more cardiac electrical signals from the sensed cardiac electrical signals. When four different cardiac electrical signals are sensed, e.g., using all four available electrodes **24**, **26**, **28** and **30** each paired with housing **15** in four different sensing electrode vectors all having one common electrode (housing **15** in this example), the remaining six possible sensing electrode vector signals may be constructed from the four sensed cardiac electrical signals. Furthermore, in some examples, a cardiac electrical signal may be constructed from one sensed and one constructed cardiac electrical signal determined from two other cardiac electrical signals or even constructed from two other constructed cardiac electrical signals.

(97) In some examples, sensing circuit **86** may include two or more sensing channels including the same or different filters, amplifiers, ADCs and/or other signal processing circuitry such that different cardiac electrical signals sensed from one or different sensing electrode vectors may be passed to control circuit **80** for processing and analysis. While sensing circuit **86** is shown having a single prefilter and amplifier **62**, ADC **63** and wideband filter **74** for passing up to four sensed cardiac electrical signals **70**, **72**, **76** and **78**, other sensing circuit configurations may include multiple channels, each having its own prefilter and amplifier, ADC and wideband filter for passing a cardiac electrical signal to control circuit **80** for storage in memory **82** and/or transmission by telemetry circuit **88**. An example sensing circuit having two sensing channels each with a prefilter

and preamplifier and ADC is generally described in U.S. Pat. No. 9,956,423 (Zhang, et al.), incorporated herein by reference in its entirety. Sensing circuit **86** may include more or fewer components than illustrated in FIG. 5 and some components may be shared between multiple sensing channels for sensing multiple cardiac electrical signals. The configuration of sensing circuit **86** as shown in FIG. 5 is therefore illustrative in nature and should not be considered limiting of the techniques described herein for sensing at least two cardiac electrical signals and constructing at least one more cardiac electrical signal. As described below, each sensed cardiac electrical and/or each cardiac electrical signal constructed from the sensed cardiac electrical signals may be analyzed according to multiple sensing threshold control parameter settings. In the illustrative examples disclosed herein, the sensed cardiac electrical signals sensed by sensing circuit **86** may be transmitted to external device **40** for post-processing and analysis for constructing at least one more cardiac electrical signal and/or determining sensed cardiac event intervals according to multiple sensing threshold control parameter settings.

(98) FIG. 6 is a diagram of a filtered and rectified cardiac electrical signal **200** illustrating one technique for adjusting the cardiac event sensing threshold **210** by sensing circuit **86** under the control of control circuit **80**. Cardiac electrical signal **200** includes an R-wave **202**, a T-wave **204**, and a P-wave **206**. The cardiac electrical signal **200** may be representative of the output of rectifier **65** received by cardiac event detector **66** as shown in FIG. 5. Alternatively, cardiac electrical signal **200** may represent a narrowband filtered and rectified signal determined from a sensed or constructed wideband filtered signal by control circuit **80** or external device processor **52** during processing and analysis of the sensed and constructed cardiac electrical signals for determining sensed cardiac events according to one or more cardiac event sensing threshold control parameter settings. Analysis performed by a processor as disclosed herein to determine sensed cardiac events according to test sensing control parameter settings mimics the processing that occurs by sensing circuit **86** for sensing cardiac events from the narrowband filtered, rectified signal by cardiac event detector **66**. In the description that follows, the operations performed to adjust the cardiac event sensing threshold **210** are described as being performed by sensing circuit **86** for the sake of convenience. However, during real time or post-processing of a cardiac electrical signal by control circuit **80**, by external device processor **52** or by another processor, it is to be understood that determination of sensed cardiac events from the cardiac electrical signal involves the same cardiac event sensing threshold adjustment operations as would be performed by sensing circuit **86** for determining when the cardiac electrical signal, sensed or constructed, is estimated to cross the cardiac event sensing threshold.

(99) In this example, the cardiac event sensing threshold **210** is an R-wave sensing threshold that is adjusted for sensing R-waves from the cardiac electrical signal **200** and determining RRI. Sensing circuit **86** adjusts the R-wave sensing threshold **210** between a starting threshold amplitude **212** and the minimum sensing threshold amplitude equal to the programmed sensitivity **220**. The starting threshold amplitude **212** may be set based on the maximum peak amplitude **203** of sensed R-wave **202**. R-wave **202** is sensed by sensing circuit **86** in response to the cardiac electrical signal **200** crossing the R-wave sensing threshold at **201**. Sensing circuit **86** produces a ventricular sensed event signal (VS) **230** (e.g., corresponding to the sensed event signal **68** output by cardiac event detector **66** shown in FIG. 5) in response to the sensing threshold crossing **201**. Sensing circuit **86** may be configured to detect the maximum peak of R-wave **202** during a post-sense blanking period **240** for determining maximum peak amplitude **203**. The starting R-wave sensing threshold amplitude **212** may be set to a percentage, e.g., between 55% and 70% or another selected percentage, of the maximum peak amplitude **203**. For example, the percentage used to set starting R-wave sensing threshold amplitude **212** can be 62.5% of the maximum peak amplitude **203**.

(100) The starting threshold **212** may be held for a sense delay interval **242** to avoid oversensing T-wave **204** as an R-wave. In other examples, the starting threshold **212** may decay at a specified decay rate for a predetermined decay interval. In the example shown, R-wave sensing threshold

210 is decreased by a step decrement **214** upon expiration of sense delay interval **242**. Sense delay interval **242** may be between 300 and 400 ms, as examples, and is 360 ms in one example. At the expiration of sense delay interval **242**, sensing circuit **86** adjusts the R-wave sensing threshold **210** from the starting amplitude **212** set to a first percentage of maximum peak amplitude **203** to an intermediate sensing threshold amplitude **216** that is a second percentage of R-wave maximum peak amplitude **203**. The second percentage is less than the first percentage. Intermediate sensing threshold amplitude **216** may be set to between 25% and 60% of the maximum peak amplitude **203** or between 30% and 35% of the maximum peak amplitude **203** as examples. Intermediate sensing threshold amplitude **216** is less than the starting sensing threshold amplitude **212** by step decrement **214**.

(101) R-wave sensing threshold **210** may be held at the intermediate amplitude **216** for a drop time interval **244** as shown in FIG. 6. In other examples, R-wave sensing threshold **210** may decay at a specified decay rate from the expiration of the sense delay interval **242** until the expiration of drop time interval **244** or until reaching the sensing floor equal to the programmed sensitivity **220**. In the example shown, upon expiration of drop time interval **244**, sensing circuit **86** adjusts R-wave sensing threshold **210** from the intermediate sensing threshold amplitude **216** to the sensitivity **220** in a step decrement **218**. The sensitivity **220** defines the minimum sensing threshold amplitude or sensing floor of R-wave sensing threshold **210**. The drop time interval **244** may be between 1 second and 2 seconds and is 1.5 seconds in one example. The sensitivity **220** may be programmable over a range of 0.075 millivolts (mV) to 1.2 mV, as examples, though lower or higher sensitivity settings may be available.

(102) Each of the first percentage used to set starting sensing threshold amplitude **212** as a percentage of maximum peak amplitude **203**, the second percentage used to set intermediate sensing threshold amplitude **216** as a percentage of maximum peak amplitude **203**, the sensitivity **220**, the post-sense blanking period **240**, the sense delay interval **242** and the drop time interval **244** may be programmable or adjustable cardiac event sensing threshold control parameters. Accordingly, processor **52** of external device **40** (and/or control circuit **80**) may be configured to determine cardiac sensed events from one or more cardiac electrical signals, sensed and/or constructed, according to one or more different settings of one cardiac event sensing threshold control parameter or combinations of different settings of two or more different sensing threshold control parameters. In illustrative examples described below, programmer **52** of external device **40** is configured to at least determine sensed cardiac events according to different settings of sensitivity **220** for at least one constructed cardiac electrical signal.

(103) As shown in FIG. 6, R-wave sensing threshold **210** is held at sensitivity **220** until the cardiac electrical signal **200** crosses the sensing threshold **210**, at sensing threshold crossing **231**, resulting in the next VS event signal **234** produced by sensing circuit **86**. It is to be understood that the cardiac electrical signal **200** may not cross the sensing threshold **210** before a pacing interval expires during some cardiac cycles. In this case, therapy delivery circuit **84** may generate and deliver a pacing pulse. At other times, the next R-wave **248** may occur earlier after R-wave **202**, before R-wave sensing threshold **210** reaches sensitivity **220** (before drop time interval **244** expires) or even before R-wave sensing threshold **210** reaches the intermediate sensing threshold amplitude **216** (before sense delay interval **242** expires). The cardiac event interval **246**, which is an RRI in this example, may be determined by control circuit **80** (or processor **52** of external device **40**) as the time interval between VS event signal **230** corresponding to threshold crossing **201** and VS event signal **234** corresponding to threshold crossing **234**.

(104) The particular behavior of R-wave sensing threshold **210** shown in FIG. 6 as it is adjusted between the starting threshold amplitude **212**, set based on the maximum peak amplitude **203**, and the sensitivity **220** is one illustrative example of how sensing circuit **86** may adjust the sensing threshold **210**. It is to be understood that a variety of cardiac event sensing threshold control parameters, e.g., R-wave sensing threshold control parameters for ventricular rate determination or

P-wave sensing threshold control parameters for atrial rate determination, may include one or more decay rates, each associated with a decay interval, and/or one or more step decrements, each associated with a drop time interval. Various cardiac event sensing threshold control parameters may be used by sensing circuit **86** for adjusting the R-wave sensing threshold **210** between the starting threshold **212** and sensitivity **220**. For example, the R-wave sensing threshold **210** may decay, linearly or non-linearly, from starting sensing threshold amplitude **212** to sensitivity **220** at a predetermined decay rate until a sensing threshold crossing or a pacing interval expires, whichever occurs first.

(105) The techniques disclosed herein for determining sensed cardiac events from one or more cardiac electrical signals (corresponding to different sensing electrode vectors) according to one or more sensing threshold control parameter settings are not limited to use with any particular sensing threshold control parameters or sensing threshold adjustment schemes. The cardiac event sensing threshold control parameters used by sensing circuit **86** to adjust sensing threshold **210**, however, are the same cardiac event sensing threshold control parameters used by control circuit **80** and/or external device processor **52** for determining sensed cardiac events during post processing of different cardiac electrical signals for determining a recommended sensing control parameter setting. While the same sensing control parameters are used, such as sense delay interval **242**, drop time interval **244** and sensitivity **220**, different settings of these sensing threshold control parameters may be used during a determination of sensed cardiac events from a sensed or constructed cardiac electrical signal. By simulating the adjustment of the cardiac event sensing threshold that is performed by sensing circuit **86** during real time cardiac event sensing, control circuit **80** or external device processor **52** can determine sensed cardiac events for multiple different combinations of sensing control parameters settings without requiring ICD **14** to be reprogrammed to the multiple different combinations and perform the real time sensing.

(106) FIG. **7** is a flow chart **300** of a method that may be performed by a medical device for determining sensed cardiac events according to multiple sensing control parameters according to one example. At block **302**, sensed cardiac electrical signals are received by a processor of a device performing the cardiac electrical signal analysis. As indicated above, the process for analyzing cardiac electrical signals as disclosed herein may be performed by a medical device that senses the cardiac electrical signals, e.g., by control circuit **80** of ICD **14** or processing circuitry of a pacemaker, cardiac monitor or other device configured to sense the cardiac electrical signals. As such, in one example, control circuit **80** receives at least two sensed cardiac electrical signals from sensing circuit **86** at block **302**.

(107) In other examples, the process for analyzing cardiac electrical signals as disclosed herein may be performed by a processor of a device that receives the sensed cardiac electrical signals from another device. For example, the processor **52** of external device **40** may receive the sensed cardiac electrical signals from ICD **14** via communication link **42**. In other examples, the processor **52** of external device **40** may receive the sensed cardiac electrical signals from an external cardiac monitoring device, e.g., cardiac monitoring device **60** shown in FIG. **1C**. In some examples, the external cardiac monitoring device **60** or the external device **40** may transmit sensed cardiac electrical signals to a patient monitoring database, such as the CARELINK® Network (Medtronic, Minneapolis, Minnesota USA), for processing and analysis by a networked computer or by a computer in a clinic, hospital or doctor's office. Depending on the processing requirements and power capacity of the device sensing the cardiac electrical signals, e.g., ICD **14**, the sensed cardiac electrical signals may be transmitted to another device for processing and analysis. In still other examples, the processing and analysis of cardiac electrical signals as disclosed herein may be performed in a distributed manner across more than one device of a medical device system. For the sake of illustration, with no limitations intended, the techniques described in conjunction with FIG. **7** and other flow charts presented herein refer to ICD **14** as the device sensing the cardiac electrical signals and external device **40** as receiving the sensed cardiac electrical signals, which may be via

cardiac monitoring device **60**, and performing subsequent processing and analysis for determining acceptable or recommended sensing control parameters.

(108) The cardiac electrical signals received at block **302** by processor **52** of external device **40** may be sensed by sensing circuit **86** of ICD **14** using at least two different sensing electrode vectors. The two sensing electrode vectors include one electrode common to both sensing electrode vectors but not both electrodes. For example, the first sensing electrode vector may include pace/sense electrodes **28** and **30** carried by lead **16** as shown in FIG. **1A** and the second electrode vector may include one of pace/sense electrodes **28** or **30** paired with housing **15** or a defibrillation electrode **24** or **26**.

(109) The two sensed cardiac electrical signals may be passed to control circuit **80** from sensing circuit **86** as raw cardiac electrical signals, e.g., as wideband filtered signals from filter **74**. The sensed cardiac electrical signals may be transmitted from ICD **14** to external device **40** as cardiac signal episodes that are a few seconds or a few minutes in duration for example. The sensed cardiac electrical signals may be received in real time via telemetry circuit **88** or stored in memory **82** and then passed to telemetry circuit **88** for transmission to external device **40** at a later time. In other examples, the sensed cardiac electrical signals may be transmitted in real time to cardiac monitoring device **60** and processed by the cardiac monitoring device or subsequently transmitted to external device **40**. In still other examples, the sensed cardiac electrical signals may be surface ECG signals received by cardiac monitoring device **60** or external device **40** as described above.

(110) At block **304**, processor **52** of external device **40** constructs a third cardiac electrical signal that is expected to be sensed from a third sensing electrode vector, different than the first and second sensing electrode vectors but including one electrode from each of the sensed cardiac electrical signal sensing electrode vectors. The third sensing electrode vector signal may be determined by determining the voltage difference between the two sensed cardiac electrical signals at each sample point.

(111) FIG. **8** is a diagram **350** of some possible sensing electrode vectors that exist between the electrodes **24**, **26**, **28** and **30** carried by lead **16** and housing **15**. Using the example given above, when the first sensing electrode vector **S1** is between the pace/sense electrode **30** and pace/sense electrode **28** and the second sensing electrode vector **S2** is between the pace/sense electrode **30** and the housing **15**, a third, constructed cardiac electrical signal that is expected to be sensed between a third sensing electrode vector **C1** (between pace/sense electrode **28** and the housing **15**) can be determined by subtracting the **S1** signal from the **S2** signal. The third cardiac electrical signal expected to be sensed by the third sensing electrode vector **C1** is therefore constructed based on two cardiac electrical signals sensed from sensing electrode vectors having one common electrode (pace/sense electrode **30** in this example).

(112) The constructed cardiac electrical signal corresponds to a sensing electrode vector **C1** including one electrode **28** from the first **S1** sensing electrode vector used to sense the first sensed cardiac electrical signal and one electrode (housing **15**) from the second **S2** sensing electrode vector used to sense the second sensed cardiac electrical signal. The constructed cardiac electrical signal corresponding to the sensing electrode vector **C1** between pace/sense electrode **28** and housing **15** is determined by determining the voltage amplitude differences of the two sensed cardiac electrical signals corresponding to vectors **S1** and **S2** at each sample point time.

(113) Referring back to FIG. **7**, in some examples, more than two sensed cardiac electrical signals may be received at block **302**, so that more than one cardiac electrical signal may be constructed from the sensed cardiac electrical signals. In the example of FIG. **1A**, all four electrodes **24**, **26**, **28** and **30** and housing **15** may be included in four different sensing electrode vectors used to sense four cardiac electrical signals so that a cardiac electrical signal expected to be sensed from each one of the remaining six possible sensing electrode vectors can be constructed. Using the four sensed cardiac electrical signals, additional cardiac electrical signals may be constructed by external device processor **52** to enable analysis of cardiac electrical signals corresponding to up to 10

different sensing electrode vectors available between the four electrodes **24**, **26**, **28**, **30** and housing **15**. For example, if four cardiac electrical signals are sensed using the four sensing electrode vectors between the housing **15** and each respective electrode **24**, **26**, **28** and **30**, the cardiac electrical signals expected from the six sensing electrode vectors (defined by electrodes **24** and **26**, electrodes **24** and **28**, electrodes **28** and **30**, electrodes **26** and **28**, electrodes **26** and **30**, and electrodes **28** and **30**) that are available between the four electrodes may be constructed from the four sensed cardiac electrical signals for a total of ten cardiac electrical signals, sensed or constructed.

(114) In some examples, a constructed cardiac electrical signal determined from two sensed cardiac electrical signals may be used by external device processor **52** to determine another constructed cardiac electrical signal. Referring again to FIG. **8**, three cardiac electrical signals may be sensed: one between the pace/sense electrodes **28** and **30** (**S1**), one between pace/sense electrode **30** and housing **15** (**S2**), and one between defibrillation electrode **26** and housing **15** (**S3**). A constructed cardiac electrical signal expected to be sensed between pace/sense electrode **28** and housing **15** (**C1**) may be determined using the first two sensed cardiac electrical signals (from sensing vectors **S1** and **S2**). Another constructed cardiac electrical signal expected to be sensed between pace/sense electrode **28** and defibrillation electrode **26** (**C2**) may be determined by determining the voltage amplitude differences between the constructed **C1** cardiac electrical signal between electrode **28** and housing **15** and the sensed cardiac electrical signal between defibrillation electrode **26** and housing **15** (**S3**). Accordingly, multiple constructed cardiac electrical signals may be determined from multiple sensed cardiac electrical signals and combinations of constructed and sensed cardiac electrical signals.

(115) At block **306**, external device processor **52** determines sensed cardiac events for at least one of the sensed and/or constructed cardiac electrical signals. For example, processor **52** may determine sensed cardiac events for the constructed cardiac electrical signal by determining a narrowband filtered and rectified signal from the constructed cardiac electrical signal and applying a cardiac event sensing threshold to the rectified signal. Processor **52** may adjust the cardiac event sensing threshold according to the same cardiac event sensing threshold control parameters that are used by cardiac event detector **66** of sensing circuit **86**. Processor **52** may repeat the analysis for at least one sensed cardiac electrical signal or both sensed cardiac electrical signals. In this way, processor **52** may generate sensed cardiac event data corresponding to a third sensing electrode vector that enables a comparison of cardiac event sensing between the third sensing electrode vector and at least one sensing electrode vector that was used to sense the cardiac electrical signals received at block **302**. This comparison of sensed events determined from the constructed cardiac electrical signal and a sensed cardiac electrical signal provides data for use in selecting the sensing electrode vector that provides reliable cardiac event sensing.

(116) In some examples, processor **52** may determine the sensed cardiac events for at least one sensed cardiac electrical signal received at block **302** in addition to determining the sensed cardiac events for at least one constructed cardiac electrical signal. Processor **52** may determine a narrowband filtered and rectified signal from each received, sensed cardiac electrical signal and apply the cardiac event sensing threshold to determine sensed cardiac events in the manner that cardiac event detector **66** (FIG. **5**) would sense cardiac events from a narrowband filtered and rectified signal. In some examples, processor **52** adjusts the cardiac event sensing threshold according to control parameters currently programmed in ICD **14**, e.g., the current post sense blanking period, percentages used to determine the starting and intermediate amplitudes of the sensing threshold, the sense delay interval, the drop time interval and the sensitivity as described above in conjunction with FIG. **6**. This analysis provides a comparison of cardiac event sensing using different sensing electrode vectors. However, processor **52** may determine sensed cardiac events for the constructed cardiac electrical signal(s) and sensed cardiac electrical signals for multiple different cardiac event sensing threshold control parameter settings.

(117) For example, processor **52** may set and adjust the R-wave sensing threshold using multiple settings for at least one of the R-wave sensing threshold control parameters to determine sensed cardiac events from each cardiac electrical signal, sensed and constructed, according to the multiple R-wave sensing threshold control parameter settings. Using sensitivity as an example, processor **52** may determine sensed cardiac events from each of the constructed and sensed cardiac electrical signals for each available sensitivity setting in ICD **14** to generate sensed cardiac event data at block **306**. Processor **52** may adjust the R-wave sensing threshold according to the sensing threshold control parameters for each cardiac electrical signal, sensed and constructed, to determine sensed cardiac events for each one of multiple sensitivity settings.

(118) In other examples, processor **52** may determine sensed cardiac event signals for each available percentage setting used to determine the starting and/or the intermediate threshold as a percentage of the maximum peak amplitude determined during the post-sense blanking period. In still other examples, processor **52** may determine sensed cardiac event signals for each available sense delay interval setting and/or each available drop time interval setting. Processor **52** may determine sensed cardiac events for a range of a given sensing threshold control parameter or different combinations of sensing threshold control parameters for each of the sensed and constructed cardiac electrical signals so that comparisons between the sensed cardiac event intervals or sensed cardiac event rate may be made between different combinations of sensing control parameters and used for identifying reliable or recommended sensing control parameters.

(119) At block **308**, processor **52** may determine a predicted sensed cardiac event rate for each sensed and constructed cardiac electrical signal for each tested sensing threshold control parameter setting. For example, processor **52** may determine a predicted sensed atrial or ventricular rate based on the determined sensed cardiac events and/or generate a series of predicted sensed cardiac event intervals, e.g., PPIs or RRI, at block **308**. It is recognized that for at least one of the sensed cardiac electrical signals and set of sensing threshold control parameters the actual time of cardiac event signals sensed by cardiac event detector **66** of sensing circuit **86** may be determined by ICD control circuit **80** and transmitted to external device **40** with the sensed cardiac electrical signals. This actual sensed cardiac event interval data, a determined cardiac event rate and/or arrhythmia detection by ICD control circuit **80** may be transmitted to external device **40** and may be used as an expected cardiac event rate for comparative analysis of other sensing control parameter settings or may be used to compare to determined sensed cardiac event intervals or rates determined from the analysis of other cardiac electrical signals according to other sensing control parameter settings. A comparison between actual sensed cardiac event data and simulated sensed cardiac event data for the same sensing control parameters may be made to verify accuracy and consistency of the determined sensed cardiac events by processor **52** for the same set of sensing control parameters. In other examples, the determination of sensed cardiac events corresponding to the programmed sensing control parameters need not be performed by external device processor **52** because the data may be available from and transmitted by ICD **14**.

(120) At block **310**, processor **52** may generate an output based on the sensed cardiac event data determined at block **308**. In some examples, the output may be generated for display on display unit **54**. For example, the output may be a graphical or tabular display of the data relating to cardiac event rate and/or cardiac event intervals determined for each sensing threshold control parameter setting applied to each sensed and constructed cardiac electrical signal. Processor **52** may generate an output of one or more recommended or acceptable sensing control parameter settings, which may be output for display, e.g., included in a GUI. A recommended or acceptable sensing electrode vector and/or sensitivity setting that results in an expected sensed cardiac event rate at a desired safety margin for sensing cardiac events may be determined and identified in a generated display, as an example. The expected or actual cardiac rate may be verified by a user or may be based on the median, mean or mode of the sensed cardiac event rates determined from all cardiac electrical signals and all sensing threshold control parameter settings used in the analysis, assuming the

actual cardiac event rate is the most frequently determined sensed cardiac event rate from among the sensing control parameters analyzed. Techniques that may be performed by processor **52** for determining a sensing control parameter that provides a desired safety margin for sensing cardiac events at the expected or actual cardiac rate are described below in conjunction with FIGS. **8-13**. In an example, the recommended sensitivity setting for a given sensing electrode vector may be one half of the minimum sensitivity setting that results in an expected or actual cardiac rate to provide a 2× (two times) safety margin for sensing cardiac events.

(121) Accordingly, processor **52** may be configured to determine a rate of the determined sensed cardiac events, determine that the rate meets expected cardiac event rate criteria and determine that an associated sensing control parameter used to determine the sensed cardiac events meeting the expected cardiac event rate criteria is a recommended sensing control parameter. The expected cardiac event rate criteria may be an actual rate (which may be defined as an acceptable rate range) determined by a user and received by processor **52** via user interface **56**. The expected cardiac event rate criteria may alternatively be a rate (or rate range) determined by processor **52** based on the most frequently determined rate from among the determined sensed cardiac events for each set of sensing control parameters (combinations of sensing electrode vector and cardiac event sensing threshold control parameter settings). In still other examples, as described below in conjunction with FIG. **9**, the expected rate criteria may be defined by an arrhythmia detection or the time to detect an arrhythmia, such as VT or VF. When a set of sensing control parameters results in determined cardiac event intervals that meet expected rate criteria as determined at block **310**, processor **52** may generate a display indicating the acceptable sensing control parameters.

(122) The output generated by processor **52** at block **310** may include a programming command that is transmitted by telemetry unit **58** to ICD telemetry circuit **88** in some examples. The programming command may include a sensing electrode vector and/or one or more cardiac event sensing threshold control parameters selected as recommended settings for providing reliable cardiac event signal sensing based on the analysis of the sensed and constructed cardiac electrical signals. Control circuit **80** responds to the programming command by setting the selected sensing electrode vector and/or sensing threshold control parameter(s). Such programming may occur without requiring user intervention. In other examples, a user may confirm a recommended setting prior to transmission of the programming command, e.g., by interacting with user interface **56** or a GUI displayed on display unit **54** or via remote monitoring from a computer or other device in communication with external device **40**.

(123) FIG. **9** is a flow chart **400** of a method performed by a medical device system for determining sensed cardiac events from multiple sensing electrode vector signals according to another example. At block **402** at least two different cardiac electrical signals are sensed by sensing circuit **86** using two different sensing electrode vectors. At least one electrode is common to both of the two sensing electrode vectors to enable control circuit **80** or external device processor **52** to construct at least one more cardiac electrical signal corresponding to a third sensing electrode vector.

(124) At block **404**, control circuit **80** may detect tachyarrhythmia based on at least one of the sensed cardiac electrical signals. The tachyarrhythmia may be an induced tachyarrhythmia or a spontaneously occurring tachyarrhythmia. In response to the tachyarrhythmia, control circuit **80** stores an episode of each of the two sensed cardiac electrical signals in memory **82** at block **406**. As described above, the two sensed cardiac electrical signals may be wideband filtered signals received from sensing circuit **86**. The stored episode extends from the first tachyarrhythmia interval detected by control circuit **80** (or earlier) until the tachyarrhythmia detection is made (or later) so that control circuit **80** or processor **52** of external device **40** is able to determine the time from the first tachyarrhythmia interval that contributed to tachyarrhythmia detection (e.g., contributed to the NID being reached) until tachyarrhythmia detection is made, referred to as the “time to detect.” While not explicitly shown in FIG. **9**, it is understood that therapy delivery circuit **84** of ICD **14** may respond to the tachyarrhythmia detection by generating a therapy, e.g., ATP or CV/DF shock

therapy.

(125) At block **408**, processor **52** constructs at least one alternative sensing electrode vector signal from the two sensed cardiac electrical signals. At block **408**, control circuit **80** may transmit the stored sensed cardiac electrical signal episodes corresponding to the detected tachyarrhythmia to external device **40** for processing and analysis. Control circuit **80** may perform the processing and analysis for determining sensed cardiac events from multiple cardiac electrical signals in some examples. However, in order to reduce processing burden and conserve power source **98** of ICD **14**, processing and analysis of sensed cardiac electrical signals may be performed by an external device, e.g., external device **40** shown in FIG. **1A**. As noted above, processing and analysis of cardiac electrical signals described in conjunction with flow charts presented herein, including the flow chart **400** of FIG. **9**, may be performed by control circuit **80**, external device processor **52**, or cooperatively by both control circuit **80** and external device processor **52** or another processing device such as a remote computer in communication with external device **40**.

(126) Accordingly, while transmission of the stored sensed cardiac electrical signal episodes is not explicitly shown in FIG. **9**, it is to be understood that transmission of sensed cardiac electrical signals and related data may occur at or after block **406** in order to enable an external processor **52** to obtain sensed cardiac electrical signals and perform subsequent processing and analysis.

Furthermore, in some examples such as when the detected tachyarrhythmia is an induced tachyarrhythmia, the sensed cardiac electrical signals may be transmitted in real time to external device **40** during the tachyarrhythmia induction procedure. When transmitted in real time, ICD telemetry circuit **88** may be capable of transmitting a greater number of sensed cardiac electrical signals (corresponding to more sensing electrode vectors) than the number of cardiac electrical signal episodes that memory **82** has the capacity to store (for later transmission). For example, four different sensing electrode vector signals may be transmitted by ICD telemetry circuit **88** in real time during a tachyarrhythmia induction procedure while two different sensing electrode vector signals may be stored in memory **82** for later transmission to external device **40**.

(127) FIG. **10** is a diagram **450** of three cardiac electrical signals that may be obtained by a processor and analyzed according to the techniques disclosed herein by a medical device system. In this example, cardiac electrical signal **452** is a sensed cardiac electrical signal that is sensed by a first sensing electrode vector, e.g., between pace/sense electrodes **28** and **30** (shown in FIG. **1A**). Cardiac electrical signal **454** is a sensed cardiac electrical signal that is sensed using a second sensing electrode vector, e.g., between pace/sense electrode **30** and housing **15**. The third cardiac electrical signal **456** is a constructed cardiac electrical signal that may be determined by processor **52** using the two sensed cardiac electrical signals **452** and **454**. Constructed cardiac electrical signal **456** corresponds to a cardiac electrical signal that would be sensed between sensing electrode **28** and housing **15**. Constructed cardiac electrical signal **456** may be determined by processor **52** by subtracting the first sensed cardiac electrical signal **452** from the second sensed cardiac electrical signal **454** (note that the millivolt scales are different for the different signals shown in FIG. **10**). By constructing a third cardiac electrical signal **456** from two sensed cardiac electrical signals **452** and **454**, processor **52** is able to determine sensed cardiac events from at least three different available sensing electrode vector signals to generate data relating to the determined sensed cardiac events, e.g., event intervals or rates, according to specified sensing threshold control parameter settings. This generated data enables processor **52** to make comparisons between sets of sensing control parameters, e.g., different combinations of sensing electrode vector and multiple sensitivities, without having to reprogram ICD **14** and without recording cardiac electrical signals sensed according to a relatively large number of different sensing control parameters. Selection of an acceptable sensing electrode vector and corresponding sensing threshold control parameter settings may be made based on the sensed cardiac event data determined by a processor configured to simulate the sensing operations of sensing circuit **86**, such as narrowband filtering and rectifying of the sensed and constructed signals and applying a multi-level cardiac event sensing threshold

amplitude to determine sensing threshold crossings.

(128) The two sensed cardiac electrical signals **452** and **454** are sensed by sensing circuit **86** during VF in this example and may be transmitted in real time to external processor **52**, e.g., during a VF induction procedure. In other examples, the two sensed cardiac electrical signals **452** and **454** are stored in memory **82** in response to control circuit **80** detecting a spontaneous VF episode and transmitted to external device **40**, e.g., in response to an interrogation command received by ICD telemetry circuit **88** from external telemetry unit **58**. In some examples, processor **52** may control display unit **54** to generate a display of sensed cardiac electrical signals **452** and **454** and the constructed cardiac electrical signal **456**, e.g., as part of a GUI that may include additional data relating to determined sensed cardiac events and associated event intervals and/or rates.

(129) The cardiac electrical signals **452**, **454** and **456** may be displayed as the wideband filtered cardiac electrical signals before narrowband filtering and rectification is performed for obtaining a filtered, rectified signal that the cardiac event sensing threshold is applied to according to selected sensing threshold control parameter settings. As shown and described below in conjunction with FIG. **11**, processor **52** may determine a filtered and rectified signal from each of the raw cardiac electrical signals **452**, **454** and **456** according to the filtering performed by ICD sensing circuit **86** prior to cardiac event detector **66** so that the sensing threshold control parameter settings applied by processor **52** for determining sensed cardiac events simulate the performance of cardiac event detector **66** for a given set of sensing control parameters (sensing electrode vector and sensing threshold control parameter settings).

(130) Returning to FIG. **9**, processor **52** determines sensed cardiac events from at least one constructed cardiac electrical signal at block **410**. In examples presented herein, processor **52** determines sensed cardiac events from multiple cardiac electrical signals, sensed and constructed, for multiple settings of at least one sensing threshold control parameter. Processor **52** may determine sensed cardiac events by identifying sensing threshold crossing times by the cardiac electrical signal under analysis when the sensing threshold is being adjusted according to a specified set of sensing threshold control parameter settings. In some examples, sensed cardiac events and associated event intervals are known for one of the sensed cardiac electrical signals transmitted to external device **40**, at least for the programmed sensing threshold control parameters that resulted in tachyarrhythmia detection. Processor **52** may determine the sensed cardiac events from the constructed cardiac electrical signal at block **410** using the same sensing threshold control parameters used by sensing circuit **86** and control circuit **80** that resulted in the tachyarrhythmia detection by control circuit **80**.

(131) Additionally, processor **52** may determine sensed cardiac events from the constructed cardiac electrical signal for multiple settings of one or more sensing threshold control parameters. For the sake of illustration, processor **52** may determine sensed cardiac events from the constructed cardiac electrical signal according to multiple sensitivity settings. Additionally, processor **52** may determine sensed cardiac events from one or both sensed cardiac electrical signals according to multiple sensitivity settings or other sensing threshold control parameter settings, which may be different than the programmed sensing threshold control parameter settings used to sense cardiac events by sensing circuit **86** that led to actual detection of the tachyarrhythmia by control circuit **80**.

(132) As indicated above, the sensed cardiac events determined as sensing threshold crossing times are determined by applying the same filtering, rectification and any other processing of the constructed cardiac electrical signal as performed by sensing circuit **86** on a sensed cardiac electrical signal prior to passing the signal to event detector **66** (FIG. **5**). Processor **52** applies selected sensing threshold control parameters in the same manner as event detector **66** to determine sensed cardiac events, e.g., ventricular sensed events or atrial sensed events, from the cardiac electrical signal under analysis. Using the example shown in FIG. **6**, processor **52** may determine a starting threshold amplitude **212** based on a maximum peak amplitude **203** determined during a blanking period **240**, hold the starting threshold **212** for a sense delay interval **242**, decrement to an

intermediate threshold amplitude **216** held for a drop time interval **244**, then decrement to the sensitivity **220** to determine sensed cardiac events (based on sensing threshold crossings) from the episode of the cardiac electrical signal under analysis. This process may be repeated multiple times using a different sensitivity setting each time to determine sensed cardiac events for multiple different sensitivity settings for a given cardiac electrical signal. The process of determining sensed cardiac events for multiple sensitivity settings may be repeated for each cardiac electrical signal, sensed or constructed.

(133) Using the determined sensed cardiac events for each cardiac electrical signal being analyzed according to each setting of one or more sensing threshold control parameters, processor **52** determines associated cardiac event intervals at block **412**, e.g., RRs or PPs, that occur between each consecutive pair of determined sensed cardiac events. Using the determined cardiac event intervals at block **414**, processor **52** may determine whether the tachyarrhythmia detection is predicted to be made from the corresponding sensing electrode vector signal and sensing threshold control parameter settings. In some examples, at block **414**, processor **52** may determine a predicted time to detect the tachyarrhythmia for each combination of sensing control parameter settings analyzed, e.g., for each combination of sensing electrode vector and cardiac event sensing threshold control parameter settings. In other examples, processor **52** may determine whether the tachyarrhythmia is predicted to be detected within a threshold interval of time, e.g., within 30 seconds, or within the time interval represented by a cardiac electrical signal episode stored in memory **82** in response to the actual tachyarrhythmia detected at block **404**.

(134) Processor **52** may determine whether tachyarrhythmia detection is expected at block **414** and determine the predicted time to detect the tachyarrhythmia. Processor **52** uses the tachyarrhythmia detection criteria used by ICD **14** relating to detecting tachyarrhythmia intervals from the determined sensed event intervals and for determining when the required NID to detect the tachyarrhythmia is reached. The time to detect the tachyarrhythmia may be determined as the time from an earliest tachyarrhythmia interval identified from among the determined sensed cardiac event intervals until a required NID is reached that includes the earliest tachyarrhythmia interval. The earliest tachyarrhythmia interval may be the first tachyarrhythmia interval identified from the cardiac electrical signal under analysis for a given set of sensing threshold control parameter settings. In other examples, the time to detect may be determined from a standardized starting time to the time the NID is reached for a given set of sensing control parameters. The standardized starting time may be the time of the first tachyarrhythmia interval of the actual sensed cardiac electrical signal that led to tachyarrhythmia detection. In other examples, the standardized starting may be the earliest or latest tachyarrhythmia interval identified from among all determined sensed cardiac event intervals for a given sensing electrode vector or all sensing electrode vectors.

(135) In some examples, processor **52** may require a minimum number of sensed cardiac events in the cardiac signal episodes for use in determining the event intervals at block **412** and determining a time to detect tachyarrhythmia at block **414**. Because tachyarrhythmia detection criteria may require a threshold number of tachyarrhythmia intervals out of the most recent predetermined number of cardiac event intervals, the number of sensed cardiac events in the received sensed cardiac signals may be required to be at least the threshold number of tachyarrhythmia intervals required to detect the tachyarrhythmia. To illustrate, if the NID is set to 30 tachyarrhythmia intervals out of the most recent 40 cardiac event intervals, processor **52** may require that at least one of the sensed cardiac electrical signal episodes include at least 30 sensed cardiac events to continue analysis of the sensed and constructed signals associated with the acquired episode. When a threshold number of sensed cardiac events are included in the sensed cardiac electrical signal(s), each of the sensed and constructed signals may undergo further analysis. In some cases, a given episode of a sensed or constructed signal may be repeated in a looping manner during the analysis at block **414** to provide a signal episode of sufficient length to predict a time to detect tachyarrhythmia. The predicted tachyarrhythmia detection time for some sensing control

parameters may be longer than the original cardiac signal episode. As such, the cardiac signal episodes may be looped to generate an episode that is up to one minute long, for example, for each sensed and constructed cardiac signal undergoing analysis to promote an adequate episode duration for predicting a tachyarrhythmia detection time according to various sensing control parameter settings.

(136) At block **416**, processor **52** may generate an output of expected tachyarrhythmia detections, the determined times to detect the tachyarrhythmia and/or related cardiac event interval or rate data for each combination of sensing control parameter settings analyzed. Processor **52** may generate data in a tabular or graphical format for display on display unit **54**, which may be part of a GUI. In some examples, processor **52** may generate an output of a recommended sensing control parameter setting, e.g., a recommended sensing electrode vector, a recommended cardiac event sensing threshold control parameter such as the sensitivity or other sensing threshold control parameter(s) described above in conjunction with FIG. **6**. As described below in conjunction with FIG. **13**, a recommended sensing threshold control parameter setting for a given sensing electrode vector may be based on the expected tachyarrhythmia detections and/or determined times to detect tachyarrhythmia for that sensing electrode vector and a desired safety margin for sensing cardiac events. In some examples, the output generated by processor **52** at block **416** includes a programming command transmitted to ICD **14** to program a recommended sensing control parameter or combination of parameters, such as a sensing electrode vector and corresponding sensitivity or other cardiac event sensing threshold control parameter(s).

(137) FIG. **11** is a diagram of a GUI **500** that may be generated as output at block **416** of FIG. **9** by external device processor **52** for display on display unit **54** according to one example. In some examples, display **54** of external device **40** is a touch-sensitive screen that is configured to both display GUI **500** to a user as well as provide touch-sensitive regions of GUI **500** that allow the user to provide input to GUI **500**. In other examples, a user may navigate to different user input portions of GUI **500**, e.g., selectable windows, pop-up-windows, menus, icons, buttons or the like, using a mouse, keyboard or other user interface input device.

(138) GUI **500** may include a display of a cardiac electrical signal **502**, a timing diagram **510**, and a data table **520**. In a cardiac electrical signal window **505** of GUI **500**, a cardiac electrical signal **502** is displayed, which may be a sensed or constructed cardiac electrical signal. In this example, cardiac electrical signal **502** is a sensed signal from a sensing electrode vector between pace/sense electrodes **28** and **30** (referred to as ring 1 and ring 2, respectively, or R1 and R2 in GUI **500**) of lead **16**. The cardiac electrical signal displayed in window **505** may be selectable by a user interacting with GUI **500**. For example, a drop down or scrollable menu **506** of available cardiac electrical signals may be included in GUI **500** to enable a user to select different cardiac electrical signals, sensed or constructed, for display in window **505**. In addition to a sensed or constructed cardiac electrical signal, GUI **500** may include a display of an electrocardiogram signal from the patient, which may be received via external ports **55** (shown in FIG. **1A**) and stored in memory **53**, to provide the user with a visual comparison between the electrocardiogram and the sensed or constructed cardiac electrical signal.

(139) Cardiac electrical signal **502** is the wideband filtered, unrectified signal received from ICD **14**. In some instances, cardiac electrical signal **502** is transmitted from ICD **14** after being stored in memory **82** by control circuit **80** in response to a tachyarrhythmia detection, e.g., a VF detection. In other instances, cardiac electrical signal **502** may be a sensed cardiac electrical signal that is transmitted in real time from ICD **14** to external device **40** during a tachyarrhythmia induction procedure. In still other examples, a constructed cardiac electrical signal determined by processor **52** during post-processing and analysis may be selected from a list in user input menu **506** for display in window **505**. In other examples, instead of displaying each sensed or constructed cardiac electrical signal one at a time in window **505**, multiple (or all sensed and constructed) cardiac electrical signals may be displayed simultaneously in window **505**. Using menu **506**, a user may be

able to select two or more cardiac electrical signals at a time for display in window **505** for comparison and review.

(140) Processor **52** may determine the narrowband filtered and rectified cardiac electrical signal **504** from wideband filtered cardiac electrical signal **502**. The narrowband filtered and rectified signal **504** may be included in the display window **505**. Filtered and rectified cardiac electrical signal **504** represents the narrowband filtered and rectified signal that would be passed to cardiac event detector **66** of sensing circuit **86** during real time cardiac event sensing by ICD **14**. Processor **52** applies the cardiac event sensing threshold, adjusted according to sensing threshold control parameter settings, to the filtered and rectified cardiac electrical signal **504** for determining sensed cardiac events. When more than one sensed and/or constructed cardiac electrical signal is selected from menu **506** for display in window **505**, the corresponding, time-aligned filtered and rectified cardiac electrical signal may be displayed along with the sensed or constructed cardiac electrical signal. In other examples, window **505** may include a display of only the narrowband filtered and rectified cardiac electrical signal(s) **504** without displaying the corresponding wideband filtered signal(s) **502** or vice versa.

(141) Timing diagram **510** includes sensed cardiac event markers **512** corresponding to sensed cardiac events determined from the filtered and rectified cardiac electrical signal **504** by processor **52**. In timing diagram **510**, each marker **512** is generated by processor **52** to indicate the time of a cardiac event sensing threshold crossing by filtered, rectified signal **504** determined by processor **52** as the time that a cardiac event would be expected to be sensed by sensing circuit **86** from the corresponding filtered and rectified cardiac electrical signal **504**. Timing diagram **510** may include multiple rows **515** of sensed cardiac event markers **512**, with each individual row displaying event makers **512** generated by processor **52** according to a different cardiac event sensing threshold control parameter setting. GUI **500** may include a drop down or scrollable menu **508** as a user input portion of GUI **500** for selecting a cardiac event sensing threshold control parameter. Using menu **508**, a user may select a control parameter from among the programmable cardiac event sensing threshold control parameters. Sensitivity is shown as the selected sensing control parameter in the menu **508** such that each individual row of rows **515** corresponds to a different sensitivity setting as displayed in a parameter setting window **516** adjacent to the respective row of sensed cardiac event markers **512**. In the example shown, the sensed cardiac event markers **512** are shown for each available sensitivity setting between 0.075 mV and 1.2 mV as listed in sensing parameter setting window **516**.

(142) In some examples, a user may select different sensing threshold control parameters from menu **508** to cause processor **52** to generate and display cardiac sensed event markers **512** for each available setting of the selected sensing threshold control parameter. For instance, a user may select the percentage used to set the starting threshold amplitude, the percentage used to set the intermediate threshold amplitude, the sense delay interval, the drop time interval or the sensitivity to cause processor **52** to generate a timing diagram **510** of rows **515** of cardiac sensed event markers **512** corresponding to each available (or user selected) setting of the selected parameter (and the cardiac electrical signal displayed in window **505**).

(143) In some cases, two or more different cardiac event sensing threshold parameters may be selected to generate a timing diagram **510** that includes a row of cardiac sensed event markers **512** for each combination of settings or a selected subset thereof with the corresponding parameter settings displayed in window **516**. In still other examples, window **516** may be a user input portion of GUI **500** to enable the user to select which settings of a selected sensing threshold control parameter are included in the data displayed in timing diagram **510** (and optionally in other portions of GUI **500**, such as table **520** described below). A user may select the sensing threshold control parameter(s) from menu **508** and select individual settings for each selected parameter from window **516**, e.g., using a mouse, pointer, touch screen or the like. In response to the user input, processor **52** may determine any data necessary from the selected cardiac electrical signal(s) and

generate a timing diagram **510** of cardiac sensed event markers **512**, which may be arranged in rows with each row corresponding to a combination of selected sensing threshold control parameter settings. While not shown in FIG. **11**, it is contemplated that GUI **500** may include a display of determined cardiac event intervals (e.g., in milliseconds) in timing diagram **510** and/or a determined cardiac event rate (e.g., in beats per minute).

(144) Timing diagram **510** may further include a tachyarrhythmia detection marker **514** in each row of rows **515** corresponding to a predicted time that tachyarrhythmia is expected to be detected according to the corresponding set of sensing control parameters. The tachyarrhythmia detection marker **514** may be aligned in time (along timeline **517** relative to the determined sensed event markers **512**) to indicate the time that an NID is reached for a given sensitivity setting (or other selected parameter setting) based on the determined sensed cardiac event markers **512**. Processor **52** may determine the time to detect tachyarrhythmia, the time to detect VF in this example, by determining and summing the time intervals between consecutive sensed cardiac event markers **512** that include the first and last VF intervals counted toward reaching the NID. Processor **52** may generate a tachyarrhythmia detection marker **514** as part of the GUI **500** displayed by display unit **54** to indicate the time that VF is detected from the filtered rectified signal **504** for each sensitivity setting **516**.

(145) In addition to or alternatively to tachyarrhythmia detection markers **514**, GUI **500** may include a window **518** indicating tachyarrhythmia is expected to be detected or not detected. Window **518** may additionally or alternatively indicate when cardiac events are expected to be sensed or not sensed at all for a given set of sensing control parameters. In the example shown, when the sensitivity is 0.45 mV or higher for sensing cardiac events from the selected cardiac electrical signal **502**, VF is not predicted to be detected. An indication of “No detection” is shown in window **518** for each sensitivity setting 0.45 mV, 0.60 mV, and 0.9 mV. When the sensitivity is set at the highest setting of 1.2 mV, cardiac events are not sensed, as indicated by no sensed cardiac event markers **512** in the row corresponding to 1.2 mV sensitivity. Instead of or in addition to indicating “No detection” in window **518**, an indication of “No sensing” may be shown in window **518** adjacent to sensitivity setting 1.2 mV.

(146) In other illustrative examples of GUI **500**, instead of showing all rows **515** of sensed cardiac event markers determined for each available or selected sensing threshold control parameter in timing diagram **510**, a user may select the sensing threshold control parameter from menu **508** and a single setting for the selected sensing threshold control parameter from window **516** (or an analogous menu of available settings). Processor **52** may generate the GUI **500** including a single row of sensed cardiac event markers corresponding to determined sensed cardiac events from the selected cardiac electrical signal. Window **516** may be configured to enable a user to scroll or toggle through different settings of the sensing threshold control parameter to visualize changes in the locations of the sensed cardiac event markers **512** and tachyarrhythmia detection markers **514** along the horizontal timeline **517** of timing diagram **510**.

(147) While not explicitly shown in GUI **500**, other user input portions of GUI **500** may include zoom in and zoom out buttons for viewing cardiac electrical signal window **505** and/or timing diagram **510** at different horizontal time resolutions and, in the case of cardiac electrical signal window **505**, vertical voltage scale resolution. Other user input portions of GUI **500** may include a pause, fast forward, reverse, store, download, save, print or other operational buttons that enable a user to view, print and/or save data displayed in GUI **500** as desired.

(148) Processor **52** may generate data included in data table **520** for display on display unit **54** in GUI **500** to summarize the sensed event data determined from each sensed and constructed cardiac electrical signal for each sensing threshold control parameter tested. In the example shown, table **520** includes the sensing electrode vector in the first column **522** that corresponds to each sensed and constructed cardiac electrical signal analyzed. The sensing electrode vectors listed in column **522** may correspond to the sensed and constructed cardiac electrical signals selectable from menu

506. In some examples, the cells in column **522**, each listing a sensing electrode vector, may be selectable by a user interacting with GUI **500** for simultaneously selecting which cardiac electrical signal is displayed in window **505** and which corresponding data is shown in timing diagram **510**. The sensing threshold control parameter **540** analyzed for determining the data displayed in timing diagram **510** may be listed in the first cell **540** of table **520**. The analyzed sensing control parameter settings may be listed in the first row **524** of table **520**. The selected sensing threshold control parameter is shown as being sensitivity in cell **540**, and the sensitivity settings analyzed are listed in the first row **524**.

(149) The data cells in the body of table **520** indicate the time (in seconds) to detect VF determined by processor **52** based on the determined sensed cardiac events for the corresponding sensing electrode vector listed in column **522** and sensitivity setting listed in row **524**. For example, the row **526** labeled Ring1-Ring2 in this example corresponds to the cardiac electrical signal **502**, sensed between pace/sense electrodes **28** and **30**. The time to detect VF in seconds listed in each data cell for each sensitivity setting corresponds to the time of the tachyarrhythmia detection marker **514** for each sensitivity setting **516** shown in the timing diagram **510**. No detection (ND) is indicated in the data cells for the sensitivity settings 0.45, 0.6 and 0.9 mV in row **526** at which VF is not detected. No sensing (NS) is indicated for the sensitivity setting 1.2 mV at which no cardiac events are predicted to be sensed as indicated by no sensed event markers for sensitivity setting 1.2 mV in timing diagram **510**.

(150) Table **520** may include the time to detect for each of the sensing electrode vectors listed in the first column **522** and analyzed by processor **52**. In this example, a second sensing electrode vector between pace/sense electrode **30** and housing **15**, referred to a “Ring1-Can” in row **528** of table **520**, and a third sensing electrode vector between the pace/sense electrode **28** and housing **15**, referred to as “Ring2-Can” in row **530** of table **520**, are shown with corresponding predicted VF detection times for sensitivity settings 0.075 to 0.3 mV. No VF detection (ND) is predicted from the sensed and constructed cardiac electrical signals corresponding to these sensing electrode vectors for sensitivity settings of 0.45 mV and higher. The higher sensitivity settings result in predicted undersensing of the fibrillation waves and a predicted failure to detect VF.

(151) The sensitivity setting and sensing electrode vector that are currently programmed in ICD **14** may be indicated in table **520**, as shown by the bolded font for time to detect corresponding to the Ring1-Ring2 sensing electrode vector and 0.9 mV sensitivity. At this sensitivity, the VF episode is not detected from the first sensed cardiac electrical signal in this example. The data cells containing a time to detect VF, corresponding to a sensing electrode vector and sensitivity setting (e.g., all sensitivity settings less than 0.3 mV for all sensing electrode vectors in this example), may be indicated as acceptable sensing control parameter settings in table **520**. Indication of acceptable sensing control parameter settings may be made in GUI **500** by generating the display of times to detect VF in the corresponding data cells in a stylized font, e.g., green or other colored font, bolded, underlined or other distinguishing characteristic. Additionally or alternatively, times to detect that correspond to unacceptable control parameter settings (all ND and NS cells for sensitivities greater than 0.3 mV in this example) may be indicated as unacceptable, e.g., by red font, strike through or other distinguishing display characteristic. Unacceptable sensitivity settings may alternatively or additionally be indicated in row **524** by shading or other formatting of the columns that include all “ND” and/or “NS” indicators.

(152) In other examples, instead of displaying the time to detect the tachyarrhythmia in the data cells of table **520**, the data cells may contain an indication of detection or no detection and/or sensing or no sensing. For example, the data cells containing a time to detect tachyarrhythmia in GUI **500** may alternatively contain a “VT” or “VF” to indicate detection of VT or VF is expected for the combination of sensing control parameters and “ND” to indicate no detection is expected. The indication of detection or no detection may distinguish combinations of sensing control parameter settings that are acceptable from sensing control parameter settings that are deemed

unacceptable due to an expected failure to detect the tachyarrhythmia episode. In some cases, a tachyarrhythmia may be detected from a sensing electrode vector signal according to a sensing threshold control parameter but at a predicted detection time that is longer than an acceptable threshold detection time. Processor 52 may generate an indication of no detection for display in GUI 500 for a specified combination of sensing control parameters that results in a predicted detection after a specified time limit, e.g., after more than 30 seconds. Processor 52 may generate an indication of detection for display in GUI 500 for a specified combination of sensing control parameters that results in detection predicted within the specified time limit, e.g., within 30 seconds or less.

(153) The GUI 500 shown in FIG. 11 illustrates data that may be included in a GUI generated for display by display unit 54. In other examples, GUI 500 may include less data or more data than shown in FIG. 11. For example, display of the cardiac electrical signal 502 and/or the narrowband filtered and rectified signal 504 may be optional. In some examples, timing diagram 510 is omitted with summary data presented in table 520 with an indication of acceptable sensing control parameter settings. In some examples, only table 520 may be shown in GUI 500 or only timing diagram 510, with or without an indication of acceptable or recommended sensing control parameter settings.

(154) A user interacting with GUI 500 may select one or more sensing control parameter settings for programming in ICD 14. For example, data cells included in table 520 may be user selectable using a pointer, mouse, touch screen or the like. A user may select a combination of sensing electrode vector and sensitivity setting by selecting a data cell in table 520 and confirming the programming selection using a user input program button or icon. In an illustrative example, upon selection by a user, a data cell 545 or its contents may become enlarged or bolded. A program confirmation user input window 546 may be displayed in response to the user selection of a data cell 545, e.g., in a programming pop-up window. The program confirmation user input window 546 may indicate the selected programmable parameters corresponding to the selected data cell, e.g., sensing electrode vector and sensitivity setting, and include “confirm” and “cancel” user input buttons for confirming or cancelling the programming of the selected settings in ICD 14. In other examples, a cell in column 522 corresponding to a specified sensing electrode vector and a cell in row 524 corresponding to a sensitivity setting may be selectable by a user for generating a sensing control parameter programming command by external device 40 for transmission to ICD 14. In response to a user input confirmation of a sensing control parameter setting(s), processor 52 may generate the programming command for transmission to ICD 14 via telemetry unit 58.

(155) In some examples, GUI 500 may include user selection buttons 542 and 544 that allow a user to select or toggle between a table view or a graph view of the data generated by processor 52. In the example shown, the table view is selected as indicated by enlarged font of button 542 and the display of table 520 in GUI 500. By clicking on the graph button 544, a user may switch the display from the table 520 to a graph of the data shown in table 520. A user may toggle back and forth between a tabular listing of the data as shown in FIG. 11 and a graphical view of the data, e.g., as shown in FIG. 12 and described below. Alternatively, a table of the generated data may appear in a pop-up window when table button 542 is clicked on or otherwise selected by a user, and a graph of the generated data may appear in a pop-up window when the graph button 544 is clicked or selected by a user.

(156) GUI 500 may include other patient-related data such as a patient name, birthdate or other identification. GUI 500 may further include other informational data such as a display of the date and time that a cardiac electrical signal episode was recorded by ICD 14 (when transmitted at a later time), therapy delivered, therapy outcome, or the like. GUI 500 may include more or fewer user input portions and/or more or fewer data windows, tables, etc. than shown in FIG. 11.

(157) The techniques set forth herein provide specific improvements to the computer-related field of programming medical devices that have practical applications. For example, the use of the

techniques herein may enable external device **40** to generate visualizations of cardiac electrical signal data, determined cardiac event data, determined cardiac event interval data, determined cardiac event rate data, and/or determined cardiac arrhythmia data corresponding to multiple sensing control parameter settings that define the cardiac event sensing performed by ICD **14**. Such visualizations may enable an external device, such as external device **40**, to inform a user as to how the ICD **14** is expected to perform in sensing cardiac events according to a variety of sensing control parameters without requiring ICD **14** to be reprogrammed to perform actual cardiac event sensing and arrhythmia detection according to the variety of sensing control parameters, which may include both different sensing electrode vectors and different sensing threshold control parameters. (158) By providing the GUI **500** or other user interface for displaying the data relating to determined cardiac events, the likelihood of human error in identifying cardiac events that are expected to be sensed by ICD **14** and in determining and programming sensing control parameters is reduced. Furthermore, the techniques disclosed herein may reduce the complexity of programming a medical device to sense cardiac events to the degree of accuracy required for such cardiac event data to be used for controlling the delivery and timing of cardiac electrical stimulation therapies, e.g., pacing and/or CV/DF therapies. As such, the techniques disclosed herein may enable a medical device, such as ICD **14**, to be programmed to sense cardiac events in a manner that is simplified, flexible, and patient-specific such that the ICD may reliably sense cardiac events to control delivery and timing of therapies.

(159) FIG. **12** is a graph **550** of time to detect data that may be determined by processor **52** and may be generated as output for display on display unit **54** according to one example. Graph **550** may be included in GUI **500** of FIG. **11** in some examples. As indicated above, a graphical display of data based on determined cardiac events for each sensing electrode vector signal may be generated instead of or in addition to a tabular display.

(160) The data represented by graph **550** represents another example of time to detect data determined by processor **52**, different than the data shown in GUI **500**. However, a similar graph could be generated and displayed in GUI **500**, e.g., by clicking on the graph button **544** of GUI **500**. In this example, the predicted time to VF detection (y-axis **552**) is determined for each available sensitivity setting (x-axis **554**) between 0.075 mV and 1.2 mV for each of two sensed cardiac electrical signals, shown by graphed lines **560** and **562**, and for one constructed cardiac electrical signal (shown by graphed line **564**). In this example, the time to detect VF from the first sensed cardiac electrical signal (S1), sensed between the pace/sense electrodes **28** and **30** (ring1-ring2) of lead **16** is plotted as the dotted line **562**. The time to detect VF from the second sensed cardiac electrical signal (S2), sensed between pace/sense electrode **30** and housing **15**, is plotted as the solid line **560**. The time to detect VF from the third cardiac electrical signal (C3), which is a constructed cardiac electrical signal expected to be sensed between the pace/sense electrode **28** and housing **15**, is plotted as the dashed line **564**.

(161) In this example, the predicted time to detect is approximately 6.5 seconds when the sensitivity is set to 0.075 mV for both the constructed C3 signal (graphed line **564**) and the sensed S1 signal (graphed line **562**). At the highest sensitivity setting of 1.2 mV, the predicted time to detect VF increases to approximately 7.8 seconds for the S1 signal (graphed line **562**) and increases to approximately 6.9 seconds for the constructed C3 signal (graphed line **564**). The predicted time to detect VF is approximately 8.8 seconds for the sensed S2 signal (graphed line **560**) when the sensitivity is 0.75 mV and increases to approximately 10.8 seconds at the highest sensitivity of 1.2 mV.

(162) Graph **550** may be generated by processor **52** as output for display in GUI **500** of FIG. **11**, which may include simultaneous or selectable displays of the corresponding cardiac electrical signals, e.g., as shown in FIG. **11**. The GUI may additionally or alternatively include a timing diagram of determined sensed cardiac event markers, tachyarrhythmia detection markers, and/or tabular listings of corresponding data as shown in FIG. **11**. As indicated above, the graph **550** may

be displayed as a pop-up window in GUI **500** in some examples.

(163) The data represented in graph **550** may be used by processor **52** to determine recommended or acceptable sensing control parameter settings. For example, a sensing electrode vector that is associated with the shortest time to detect tachyarrhythmia may be selected as a recommended electrode vector. In another example, a sensing electrode vector corresponding to tachyarrhythmia detection at all analyzed sensitivity settings may be identified as an acceptable sensing electrode vector. A recommended sensing electrode vector may be identified as a sensing electrode vector with all times to detect for all analyzed sensitivity settings being within a threshold difference of each other or less than a threshold detection time limit. Conversely, when a sensing electrode vector is associated with a longest time to detect, or even a predicted failure to detect the tachyarrhythmia, the sensing electrode vector may be rejected for use as a sensing electrode vector or indicated as a non-acceptable or non-recommended sensing electrode vector. Other sensing control parameters, e.g., sensing threshold control parameters used for setting and adjusting the cardiac event sensing threshold, may be determined as acceptable or recommended settings based on the data represented in FIGS. **11** and **12**.

(164) In the example of FIG. **12**, graph **550** is shown as a plot of time to VF detection (y-axis **552**) as a function of sensitivity (x-axis **554**). It is to be understood that the predicted time to detect an arrhythmia may be plotted as a function of a different sensing threshold control parameter when a different control parameter is selected to be analyzed. While only three cardiac electrical signals are represented in the graph **550**, any number of cardiac electrical signals, sensed and/or constructed, analyzed by processor **52** may be represented in graph **550**. A user interacting with the GUI may select which cardiac electrical signals are represented in the graph. In some examples, a user may select a sensing control parameter for programming in ICD **14** by clicking on a sensitivity setting and/or a sensing electrode vector that is displayed in the GUI that includes graph **550**.

(165) While graph **550** is shown as a plot of times to detect VF, it is contemplated that other data determined by processor **52** may additionally or alternatively be represented in a graph that is generated as part of a GUI. For example, determined cardiac event rates or cardiac event intervals may be plotted as a function of a sensing control parameter. The rate or intervals may be determined as the mean, median, maximum, minimum or other metric of the RRI's or PPI's that are determined from each analyzed cardiac electrical signal for each respective sensing threshold control parameter setting (or combinations of parameter settings).

(166) FIG. **13** is a flow chart **600** of a method that may be performed by external device processor **52** (or control circuit **80**) for determining a recommended sensing control parameter setting according to one example. At block **602** of FIG. **13**, processor **52** may determine the expected times to detect a tachyarrhythmia from a cardiac electrical signal, sensed or constructed, corresponding to each sensing electrode vector under analysis and each sensing threshold control parameter setting (or combinations of sensing threshold control parameter settings). Using the illustrative examples of the two different data sets represented in FIGS. **11** and **12**, processor **52** may determine the time to detect VF for each available sensitivity setting from each of at least two sensed cardiac electrical signals, sensed using different sensing electrode vectors, and at least one constructed cardiac electrical signal, constructed from the two sensed cardiac electrical signals.

(167) The times to detect may be determined by determining sensed cardiac events for each combination of sensing electrode vector and sensing threshold control parameter(s), determining the associated expected sensed cardiac event intervals, identifying tachyarrhythmia intervals among the sensed cardiac event intervals, counting the tachyarrhythmia intervals to determine when the NID is reached for detecting the tachyarrhythmia, and summing all determined cardiac event intervals between the first tachyarrhythmia interval and the last tachyarrhythmia interval that contributed to the NID being reached. It is to be understood that, when a given set of sensing control parameters are programmed into ICD **14**, the actual tachyarrhythmia detection by ICD **14** may be at the predicted time of the NID being reached or later due to processing delays. For

instance, when additional detection criteria such as morphology-based detection criteria, noise rejection criteria or other criteria that require additional cardiac electrical signal analysis and processing are required by control circuit **80** in order to determine that all detection criteria are met before detecting the tachyarrhythmia, an actual tachyarrhythmia detection time may be later than a predicted time to NID being reached.

(168) At block **604**, processor **52** may determine a time to detect limit for each sensing electrode vector. In one example, the time to detect limit may be determined as the longest time to detect that is within a predetermined time interval of the shortest time to detect for a given sensing electrode vector. The predetermined time interval may be between 1 and 7 seconds, as examples, and is 2.5 seconds in one example. Using the data of FIG. **12** as an example, the shortest time to detect for the constructed cardiac electrical signal (C3) shown by graphed line **564** is approximately 6.5 seconds at sensitivity settings from 0.075 mV and less than 0.3 mV. The longest time to detect is approximately 6.9 seconds at sensitivity settings from 0.3 to 1.2 mV. Processor **52** may determine the time to detect limit as 2.5 seconds greater than the shortest time to detect, in this case 6.5 seconds plus 2.5 seconds or 9 seconds. All times to detect for the C3 sensing electrode vector fall within this time to detect limit. The time to detect limit may be determined as the detection time at the most sensitive sensitivity setting, e.g., 0.075 mV plus a predetermined time interval, e.g., 2 to 5 seconds or about 2.5 seconds as examples.

(169) At block **606** of FIG. **13**, processor **52** may identify the cardiac event sensing threshold control parameters that correspond to the time to detect limit. In the example of FIG. **12**, processor **52** may identify the greatest sensitivity setting that resulted in the time to detect being at or less than the time to detect limit. This greatest (highest value) sensitivity setting corresponds to the lowest sensitivity of sensing circuit **86** in sensing cardiac events (highest sensing floor) that still results in the tachyarrhythmia detection within the time to detect limit. Continuing with the example given above, the greatest sensitivity setting for the C3 sensing electrode vector that results in a time to detect that is within 2.5 seconds of the shortest time to detect is 1.2 mV because the 6.9 seconds time to detect is less than the 9 second time to detect limit.

(170) At block **608**, processor **52** determines the cardiac event sensing threshold control parameter setting that provides a safety margin for detecting the tachyarrhythmia relative to the threshold control parameter setting identified at the time to detect limit. Using the same example of the C3 graphed line **564** in FIG. **12**, the greatest sensitivity setting within the time to detect limit is 1.2 mV. A lower sensitivity setting may be determined at block **608** that provides a desired safety margin for sensing cardiac events, which may be a factor of the greatest sensitivity setting. For example, if a 2× safety margin for sensing cardiac events is desired, the sensitivity setting that is half of the greatest sensitivity setting associated with a time to detect within the time to detect limit may be determined at block **608**. In this example, a sensitivity setting of 0.6 mV (half of 1.2 mV) is the recommended sensitivity setting for the sensing electrode vector associated with the constructed C3 signal represented by graphed line **564** to provide a 2× safety margin. When a 3× safety margin is desired, processor **52** may determine 0.45 mV as the recommended sensitivity setting. When a 4× safety margin is desired, processor **52** may determine 0.3 mV as the recommended sensitivity setting for the C3 sensing electrode vector.

(171) With continued reference to FIG. **12**, the time to detect limit corresponding to the S1 sensing electrode vector (graphed line **562**) is approximately 9 seconds (using the predetermined time interval of 2.5 seconds plus the shortest time to detect of 6.5 seconds at a sensitivity of 0.75 mV). The longest time to detect for the S1 sensing electrode vector is 7.8 seconds, which is within the time to detect limit. The greatest sensitivity setting resulting in a time to detect within the time to detect limit is 1.2 mV. Processor **52** may determine the recommended cardiac event sensing threshold parameter at block **608** of FIG. **13** for the associated S1 sensing electrode vector as the sensitivity setting of 0.6 mV for a 2× safety margin for sensing cardiac events.

(172) The time to detect limit for the S2 sensing electrode vector associated with graphed line **510**

in FIG. 12 is approximately 11.1 seconds, based on an acceptable increase of 2.5 seconds from the shortest time to detect of approximately 8.6 seconds at 0.075 mV. The longest time to detect within this time to detect limit is identified by processor 52 as approximately 10.8 seconds at the 1.2 mV sensitivity setting, which is within the time to detect limit. Processor 52 may determine the recommended sensitivity setting as being 0.6 mV for a 2× safety margin for sensing cardiac events.

(173) In these examples, processor 52 determines the time to detect limit as the time to detect at the lowest sensitivity setting, e.g., 0.075 mV in the example of FIG. 12, plus an acceptable increase in time to tachyarrhythmia detection, e.g., 2.5 seconds greater than the time to detect at the lowest sensitivity setting. Note that the lowest value of the sensitivity setting results in the highest sensitivity for sensing cardiac events because the cardiac event sensing threshold is decreased to the lowest possible amplitude (sensing floor), e.g., after the drop time interval as shown in FIG. 6. In other examples, the time to detect limit may be based on the shortest time to detect determined for all sensing electrode vectors plus an acceptable increase in time to detection. For example, the time to detect limit for all sensing control parameters being analyzed may be set to a predetermined interval (e.g., 2 to 7 seconds) greater than the shortest time to detect, approximately 6.5 ms for the constructed C3 signal (graphed line 564) at a sensitivity of 0.075 mV. In some examples, if the maximum time to detect for a given sensing electrode vector is greater than the time to detect limit, processor 52 may reject that sensing electrode vector in determining recommended or acceptable settings for sensing control parameters at block 610. Processor 52 may determine the time to detect limit as a percentage increase of the shortest time to detect for a given sensing electrode vector or out of all sensing electrode vectors in other examples. In still other examples, the time to detect limit may be a predetermined time interval, e.g., 12 seconds or less, 15 seconds or less, 24 seconds or less, or 30 seconds or less. The greatest sensitivity setting that results in a time to detect that is within the time to detect limit for a given sensing electrode vector may be used by processor 52 for determining a recommended sensitivity setting according to a desired safety margin at block 608.

(174) In another example with reference to the example data represented in table 520 of FIG. 11, processor 52 may determine the shortest time to VF detection as 7.56 seconds for 0.075 mV sensitivity from the cardiac electrical signal sensed using the ring1-ring2 sensing electrode vector (row 526). The highest acceptable increase in the time to detect may be determined as 7.56 seconds plus an acceptable increase of 2.5 seconds for a time to detect limit of approximately 10 seconds. The greatest sensitivity setting that results in a VF detection time within the time to detect limit is 0.2 mV, with a time to detect of 8.32 seconds (11.34 seconds for sensitivity setting 0.3 mV exceeds the time to detect limit). Processor 52 may determine the 0.1 mV sensitivity setting as the recommended setting to provide a 2× safety margin (half of 0.2 mV) for sensing cardiac events using the Ring1-Ring2 sensing electrode vector.

(175) For the second sensing electrode vector (Ring1-Can, row 528), the time to detect limit is 11.37 seconds (2.5 seconds plus the 8.87 seconds time to detect at sensitivity setting 0.075 mV). The greatest sensitivity setting falling within the time to detect limit is 0.3 mV, so the recommended setting for a 2× safety margin is 0.15 mV. For the third sensing electrode vector (Ring2-Can, row 530), the time to detect limit is 10.20 seconds (2.5 seconds plus 7.70 seconds at the 0.075 sensitivity setting). The greatest sensitivity setting resulting in a time to detect within this time to detect limit is 0.2 mV, so that the recommended sensitivity setting for a 2× safety margin is 0.1 mV for the third sensing electrode vector 530. It is to be understood that a different predetermined time interval greater than or less than 2.5 seconds may be used for determining the time to detect limit and other safety margins may be used to identify a recommended sensitivity setting for a given sensing electrode vector.

(176) Returning to FIG. 13, at block 610, processor 52 may generate an output of the recommended sensing control parameter settings determined at block 608. As described above, the output may be a programming command transmitted to ICD 14, which may include a recommended sensing electrode vector and/or sensitivity setting (and/or one or more other sensing threshold control

parameters). Additionally or alternatively, the output at block **610** may include generating a display of recommended or acceptable sensing control parameter settings in a GUI on display unit **54**. For example, GUI **500** of FIG. **11** may indicate the recommended sensing threshold control parameter setting(s) for each sensing electrode vector that is analyzed by highlighting the recommended setting by distinct color, size, underlining or other stylized font or shading or color of a data cell in table **520** or a data point in graph **550**, as examples.

(177) The process of FIG. **13** may be repeated each time a tachyarrhythmia induction is performed and/or each time external device **40** interrogates ICD **14** and retrieves sensed cardiac electrical signal episodes corresponding to detected tachyarrhythmia episodes. Alternatively, the process of FIG. **13** may be performed upon command by a user interacting with user interface **56**. For example, a user may select sensing electrode vectors and sensing threshold control parameters to be evaluated via GUI **500** and initiate a tachyarrhythmia induction. Processor **52** may receive the sensed cardiac electrical signals from ICD **14** during the tachyarrhythmia induction and process the sensed and constructed cardiac electrical signals for generating the determined cardiac events and data presented in GUI **500**.

(178) As indicated above, control circuit **80** of ICD **14** may be configured to perform some or all of the techniques disclosed herein. In that case, the process of FIG. **13** may be performed by control circuit **80** after detecting a tachyarrhythmia episode without waiting for communication with an external device. At block **610**, control circuit **80** may adjust a sensing control parameter to a recommended setting determined at block **608**. For example, if the sensitivity setting corresponding to a desired safety margin (which may be programmable) changes according to the determined times to detect for the currently selected sensing electrode vector, control circuit **80** may adjust the sensitivity setting at block **610** to the recommended setting determined at block **608**. If a sensing electrode vector is determined to have a prolonged time to detect compared to other sensing electrode vectors (or is predicted to fail to detect the tachyarrhythmia at one or more sensing threshold control parameter settings) according to the post processing analysis, control circuit **80** may select a different sensing electrode vector at block **610**.

(179) FIG. **14** is a GUI **650** that may be generated as output by processor **52** for display on display unit **54** according to one example. GUI **650** may be generated by external device **40** for visualization of the performance of cardiac event sensing by ICD **14**. GUI **650** may include a cardiac electrical signal window **651** that may display the cardiac electrical signal corresponding to a sensing electrode vector, which may be selectable by a user via a drop down or scrollable menu **653**. GUI **650** may include a table **655** listing programmable sensing threshold control parameter settings, shown as sensitivity settings **652** in this example, and a corresponding time to detect tachyarrhythmia **656** as determined by processor **52**, e.g., using the techniques described in conjunction with FIG. **9**, for each of the sensitivity settings **652**. The data shown in table **655** corresponds to the sensing electrode vector signal, sensed or constructed, selected in cardiac electrical signal window **651**. It is to be understood that processor **52** may generate a table of data or tables of data that include the predicted time to detect the tachyarrhythmia, e.g., time to reach NID, for each sensitivity setting (or other sensing threshold control parameter setting or combinations of settings) for multiple sensing electrode vectors, sensed and constructed.

(180) In this example, the time to detect limit is determined based on the time to detect at the lowest sensitivity setting of 0.075 mV (corresponding to highest sensitivity for sensing cardiac event signals). Processor **52** determines the time to detect limit by adding an acceptable increase to the time to detect at 0.075 mV sensitivity, e.g., by adding 2.5 seconds. The time to detect limit is 9.7 seconds in this example (7.2 seconds plus 2.5 seconds). Processor **52** may determine the greatest sensitivity setting at which the time to detect is equal to or less than the time to detect limit. In this example, the times to detect at sensitivity settings 0.9 mV and 1.2 mV (9.8 seconds and 11 seconds, respectively) are greater than the time to detect limit of 9.7 seconds. Table **655** may include a safety margin column **654** indicating the safety margin for each sensitivity setting as

determined based on the greatest sensitivity setting that results in a time to detect that is less than or equal to the time to detect limit. When a sensitivity setting results in a time to detect that is greater than the limit, processor **52** may determine a $0\times$ safety margin **662** for that sensitivity setting, as shown for sensitivity settings 0.9 and 1.2 mV. The GUI **650** generated by processor **52** may also denote these unacceptable sensitivity settings by demarcating these sensitivity settings 0.9 and 1.2 from other sensitivity settings by a dashed line (as shown), distinct color, size, or other stylized font, shading or other distinguishing display characteristics.

(181) In the example shown, processor **52** identifies 0.6 mV as the greatest sensitivity setting that results in a time to detect of 7.6 seconds that is equal to or less than the time to detect limit of 9.7 seconds. Based on this sensitivity setting of 0.6 mV, the safety margin of each of the other sensitivity settings less than 0.6 mV may be determined by processor **52** and displayed in safety margin column **654**. The safety margins displayed in column **654** are determined by dividing the greatest sensitivity setting 0.6 mV corresponding to a time to detect **658** within the time to detect limit by the lower sensitivity setting. In this example, 0.6 mV has a $1\times$ safety margin, 0.3 mV has a $2\times$ safety margin, 0.2 mV has a $3\times$ safety margin and so on.

(182) Processor **52** may determine a recommended sensitivity setting **660** that is a minimum desired safety margin for sensing cardiac electrical signals, e.g., a $2\times$ safety margin. The safety margin may be rounded up to a whole number as shown. When a minimum $2\times$ safety margin is programmed or desired for sensing cardiac electrical signals, the recommended sensitivity setting **660** is 0.3 mV in the example shown, half of 0.6 mV. As mentioned above, a table similar to table **650** may be generated by processor **52** and displayed on display unit **54** for each sensing electrode vector under analysis or a combined table displaying the sensitivity settings and corresponding safety margins and times to detect determined for each sensing electrode vector may be generated and displayed. The recommended sensing threshold control parameter, in this case sensitivity, for each sensing electrode vector may be highlighted in the table, e.g., by stylized font, highlighted by colored (e.g., green) or shaded cells in table **655**, encircled by a border as shown in table **650** or the like.

(183) In some examples, acceptable sensing threshold control parameter settings, e.g., all settings resulting in at least a $2\times$ (or other specified) safety margin may be highlighted by color, bolding, or other stylized font, colored fill of cells in table **655** or other formatting options in addition to highlighting the recommended sensing threshold control parameter **660**. For example, the rows corresponding to less than 0.3 mV sensitivity having a predicted safety margin for sensing cardiac events of at least $2\times$ may be highlighted as acceptable sensitivity settings, e.g., in green or yellow font or cell fill. The recommended setting of 0.3 mV at the $2\times$ safety margin may be highlighted by green font or green cell fill and/or enlarged, bolded or encircled to be distinguished as the recommended setting.

(184) Additionally or alternatively, settings that are not considered acceptable, e.g., any setting with a $1\times$ safety margin or less, may be shaded or grayed out, shown in red or displayed with other formatting of the table cells and/or font of the cell contents to indicate settings that are not recommended or not acceptable. In another example, all rows of table **655** corresponding to a $2\times$ safety margin or more (0.3 mV sensitivity or less) may be shown in green. The recommended setting **660** at the $2\times$ safety margin may be enlarged, bolded, encircled or otherwise formatted to stand out among the acceptable settings. The sensitivity settings that result in a $1\times$ safety margin may be indicated as unacceptable settings, e.g., by displaying rows corresponding to 0.45 mV and 0.6 mV sensitivity in yellow. The rows corresponding to a time to detect beyond the time limit (or any rows corresponding to a predicted failure to detect the tachyarrhythmia) may be displayed in red, grayed out, or otherwise formatted to indicate that these settings (0.9 and 1.2 mV sensitivity in this example) are not recommended.

(185) In the various GUIs described and shown herein, sensing control parameter settings, cardiac electrical signals corresponding to a sensing electrode vector, timing markers, tables, graphs, or

other representations of determined sensed cardiac events and/or predicted tachyarrhythmia detections included in the display of the GUI may be formatted according to various formatting schemes that distinguish between acceptable, recommended and/or not recommended or not acceptable sensing control parameter settings. Such formatting schemes may include colors (e.g., green, yellow and red), shading, bolding, size or other formatting options to provide a visual representation to a user that readily discerns a recommended setting and/or acceptable settings from other settings of sensing control parameters. When a setting is acceptable or recommended, the displayed setting may be selectable by a user, e.g., by clicking on row **660**, to initiate programming of the acceptable or recommended setting. When a setting is not acceptable or recommended, the display setting may not be selectable by a user. When the setting is clicked on or touched by a user in interacting with GUI **650**, a warning or error message may occur indicating that the selection is not acceptable or recommended.

(186) While table **655** is shown displaying times to detect tachyarrhythmia in column **656**, other data that is based on the determined sensed cardiac events may be displayed in table **655**, in addition to or instead of times to detect. For example, a column of table **655** may indicate whether tachyarrhythmia detection is made or not, which may be required to be within some maximum time limit, e.g., using the annotations VF, VT, or ND as examples. When no cardiac events are sensed according to a sensing control parameter setting, the indication “NS” may be included in table **655**. In other examples, table **655** may include a column indicating a cardiac rate and/or cardiac event intervals determined by processor **52** from the selected cardiac electrical signal, which may be selected and displayed in window **651**.

(187) FIG. **15** is a diagram of a GUI **700** that may be generated as output by processor **52** for display on display unit **54** according to another example. GUI **700** may include a user input portion **702** to enable a user to start a tachyarrhythmia induction, cancel the tachyarrhythmia induction, retrieve sensed cardiac electrical signals from ICD **14**, which may be stored by ICD **14** in response to detecting spontaneous tachyarrhythmia episodes, and select a therapy programming screen for programming tachyarrhythmia therapies.

(188) GUI **700** may include a sensing parameter window **704** for listing programmable sensing control parameters and corresponding test settings selected for generating determined sensed event data and currently programmed settings stored in ICD **14**. The test settings may be user inputs that enable the user to select which setting(s) of each programmable sensing control parameter is evaluated by processor **52** in determining sensed cardiac events and corresponding event intervals, rates and detected arrhythmias. In some examples, a user may input which sensing electrode vector signals are analyzed. For instance a user may select which signals are sensed by ICD **14** for transmission to external device **40** and/or which signals are constructed by selecting the vectors via a user input in sensing parameter window **704**.

(189) GUI **700** may include informational data **706** displayed as chiclets, icons, windows, or the like for displaying the date and time, providing user selectable patient information, patient heart rate or other informational data. GUI **700** may include other user input chiclets or icons, such as the user inputs **740** that enable a user to print data, print a cardiac electrical signal test strip, undo an action, or enable external device **40** to transmit a programming command, as a few examples.

(190) GUI **700** may further include a data table **710**, a cardiac electrical signal window **720**, and/or data graph window **730**. Table **710** may include data and information relating to the determined sensed cardiac events, event intervals, rates and/or predicted tachyarrhythmia detection determined by processor **52** upon processing selected cardiac electrical signals, constructed and/or sensed, according to selected sensing threshold control parameter settings. The cardiac electrical signal window **720** may display the cardiac signal episodes for each selected sensing electrode vector. User inputs included in the cardiac electrical signal window **720** may allow the user to move forward, backward, pause, freeze, increase or decrease the vertical scale or other adjustments to the display of the cardiac electrical signals.

(191) One or more graphs may be displayed in data graph window **730**. The graphs in window **730** may represent data determined by processor **52** for more than one tachyarrhythmia episode, induced or spontaneous. In the example shown, graphs of the predicted time to detect (time to NID met) for two different VF episodes each include all predicted times to detect for all available sensitivity settings for each of three different sensing electrode vectors,

(192) Features of the illustrative GUI **700** and other example GUIs illustrated or described herein may be combined in any combination or arrangement and are not limited to the combinations and arrangements shown here. Furthermore, a GUI presenting sensing control parameter settings and data may include multiple screens and windows that may be toggled between on display unit **54**.

(193) FIG. **16** is another example of GUI **750** that may be displayed by external device **40** to present a visualization of sensed cardiac event related data determined by processor **52**. GUI **750** may include elements described above in conjunction with GUI **700** of FIG. **15**. Reference numbers in FIG. **16** correspond to identically numbered elements shown in FIG. **15**. GUI **750** may include a table of sensing control parameter settings **760** with an indication of safety margin and recommended sensing control parameter setting, as generally described above in conjunction with FIG. **14**. In this example, a user input **761** may be included to allow a user to select which sensing electrode vector data is displayed in table **760**. The selected sensing threshold control parameter settings, e.g., sensitivity settings, may be listed with corresponding safety margins for sensing cardiac events as determined by processor **52**. The safety margins may be determined according to the methods described in conjunction with FIG. **13**. In the example of FIG. **16**, table **760** includes an indication (yes or no) as to whether tachyarrhythmia is expected to be detected by ICD **14** using the selected sensing control parameters (instead of listing the time to detect as shown in FIG. **14**). In the example shown, tachyarrhythmia detection is expected for all sensitivity settings, but the time to detect at sensitivities 0.9 mV and 1.2 mV is longer than the time to detect limit resulting in a 0× safety margin being displayed.

(194) A user may select different sensing electrode vectors (sensed or constructed) from a user input menu **761** for selecting which safety margin and tachyarrhythmia detection data is displayed. A user may accept the recommended sensitivity setting for the selected sensing electrode vector (as indicated by dashed box in table **760**) by clicking on the program button in user inputs **740**.

(195) FIG. **17** is a flow chart **800** of a method for determining acceptable or recommended sensing control parameters according to another example. For convenience, the method of flow chart **800** is described in conjunction with external device processor **52** performing the processing and analysis though other processors included in a medical device system may perform all or portions of the method of FIG. **17**. The process of flow chart **800** may be performed to generate data for display in a GUI to enable visualization of the performance of ICD in sensing cardiac events according to various sensing control parameters.

(196) At block **802**, a user interacting with a GUI displayed on display unit **54** may enter selected test sensing control parameters and may specify which ones of available programmable settings are to be tested for each selected sensing control parameter. External device processor **52** receives the user input test parameter settings, e.g., via the GUI or another user input interface device. For example, as shown in FIGS. **15** and **16**, a user may select test settings in window **704** for one or more of the sensing electrode vector, post-sense blanking period, starting sensing threshold amplitude percentage of maximum peak, post-sense blanking period, sense delay interval, intermediate sensing threshold amplitude percentage of the maximum peak, drop time interval and/or sensitivity (all described above in conjunction with FIG. **6**).

(197) Sensitivity has been primarily described in the illustrative examples presented herein as the sensing threshold control parameter that is evaluated at multiple programmable settings for each one of the selected sensing electrode vectors. Sensitivity is a key sensing control parameter that can directly and significantly impact ICD sensitivity to detecting VF and providing necessary CV/DF therapy. However, other sensing threshold control parameters can impact the accuracy of cardiac

event sensing and the sensitivity of ICD **14** in detecting VF or other arrhythmias. For example, different settings of other sensing threshold control parameters described herein may result in different determined sensed cardiac events due to oversensing of P-waves, T-waves, non-cardiac noise (muscle noise, electromagnetic interference, or other noise artifacts), or double counting of the QRS complex or undersensing of R-waves.

(198) While selection and programming of sensitivity (and associated sensing electrode vector) may be a high priority in promoting safe and effective detection and treatment of VT/VF, selecting and programming the settings of other sensing threshold control parameters may address and improve cardiac event sensing performance of ICD **14** to avoid oversensing and/or undersensing of cardiac events according to patient-specific needs. For example, ICD sensing circuit **86** may oversense P-waves in some patients, which could lead to false VT/VF detection. In order to avoid P-wave oversensing, the drop time interval and/or the intermediate amplitude of the R-wave sensing threshold may be analyzed at multiple settings. In other patients, ICD sensing circuit **86** may oversense T-waves, e.g., due to a patient-specific long QT interval, requiring analysis for selecting the sense delay interval and/or the intermediate amplitude settings for controlling the R-wave sensing threshold. Sensitivity and sensing electrode vector settings may both need to be analyzed for identifying acceptable settings when oversensing of non-cardiac noise is occurring or R-waves are relatively low amplitude on a selected sensing electrode vector.

(199) At block **804**, processor **52** receives at least two sensed cardiac electrical signals from ICD **14**. Processor **52** may construct at least one alternative sensing electrode vector signal using the two sensed cardiac electrical signals at block **806**. As described above, multiple sensed cardiac electrical signals may be used to construct one or more additional cardiac electrical signals for evaluating performance of different sensing threshold control parameters in sensing cardiac events.

(200) At block **808**, processor **52** may determine the sensed cardiac events from each cardiac electrical signal selected for analysis according to a first sensing threshold control parameter. One or more settings of the first sensing threshold control parameter may be applied to each signal to determine the effect of different settings on cardiac event sensing and the resulting sensed cardiac event intervals (determined at block **810**) and rate. In some examples, all available programmable settings of the first sensing threshold control parameter may be applied or a subset of settings that may be selected by the user. In some examples, the first sensing threshold control parameter tested at block **808** is any of the sensing control parameters listed above except sensitivity. Accordingly, the percentage used to set the starting threshold, the percentage used to set the intermediate threshold, the sense delay interval or the drop time interval may be the first sensing threshold control parameter evaluated at block **808**. Because sensitivity significantly affects tachyarrhythmia detection in most patients, sensitivity may be evaluated after evaluating another sensing threshold control parameter that may need to be selected based on cardiac signal analysis for an individual patient due to noise, oversensing or other sensing issues, which may be patient specific.

(201) At block **812**, processor **52** determines a setting of the first sensing threshold control parameter that results in the determined cardiac events meeting expected rate criteria. The expected rate criteria may be a heart rate confirmed by a user or determined from an electrocardiogram input. In other examples, the expected rate criteria may be a tachyarrhythmia detection, which may be required to occur within a specified time limit. In some cases, when oversensing is an issue, a setting of the first sensing threshold control parameter may result in a determined sensed cardiac event rate that is faster than an expected rate. In other instances, undersensing may occur resulting in a determined sensed cardiac event rate that is slower than the expected rate.

(202) When the determined cardiac event intervals for a first sensing threshold control parameter setting meet expected rate criteria, indicating that a possible oversensing or undersensing issue has been resolved, the setting identified at block **812** may be applied to the respective cardiac electrical signal being evaluated to redetermine sensed cardiac events according to each of the available sensitivity settings at block **814**. When a sensing threshold control parameter other than sensitivity

needs to be evaluated for a given patient, sensed cardiac events may be redetermined for all sensitivity settings (block **814**) by processor **52** using the new setting of the first sensing threshold control parameter to determine the sensitivity setting that meets a desired safety margin (block **816**) according to the techniques described above.

(203) At block **818**, processor **52** may generate an output, e.g., data for display in a GUI for visualization by a user and/or a programming command. The data representing the determined sensed cardiac events, determined sensed cardiac event intervals or rate, detection of a tachyarrhythmia and/or time to detect for each of the first sensing threshold control parameter settings may be presented in the GUI, similar to the GUIs described above. Additionally or alternatively, the first sensing threshold control parameter setting identified at block **812** may be indicated as the recommended setting and the data corresponding to testing different sensitivity settings using the recommended first sensing threshold control parameter setting may be presented in the GUI. The GUI enables a user to select and program an acceptable sensitivity, e.g., according to a desired safety margin, for sensing cardiac events when the first sensing threshold control parameter setting is at the recommended value.

(204) In an illustrative example, ICD **14** may be oversensing P-waves in a patient requiring the drop time interval be increased to avoid P-wave oversensing. As shown in FIG. **15**, a user may select a test drop time interval of 2.0 seconds when the programmed drop time interval is set at 1.5 seconds. Processor **52** may determine sensed cardiac events for all sensitivity settings at the new drop time interval of 2.0 seconds and generate a table, graph and/or timing diagram representing the determined sensed cardiac event data as described above. Processor **52** may determine the recommended sensitivity setting for achieving a desired safety margin for sensing cardiac events when the new drop time interval is applied.

(205) For the sake of convenience, the flow chart **800** depicts determining sensed cardiac events from each cardiac electrical signal according to selected settings of the first sensing threshold parameter and subsequently determining sensed cardiac events according to each sensitivity setting. It is to be understood, however, that disclosed operations may be performed in a different order than shown in the flow charts presented herein or in parallel operations. For example, processor **52** may determine sensed cardiac events for all combinations of sensing control parameters (each sensing electrode vector, each setting of the first sensing threshold control parameter, and each setting of sensitivity). A user, interacting with a GUI such as the GUI **750** of FIG. **16** or other examples presented herein, may select the test setting of the first sensing threshold control parameter and the sensing electrode vector. Processor **52** may then generate the display representative of the determined sensed cardiac event data for all sensitivity settings and the selected sensing electrode vector and first sensing threshold control parameter setting. In this way, a user interacting with GUI **750** may switch between different combinations of sensing electrode vector and the first sensing threshold parameter setting to determine a combination and associated sensitivity setting that results in a desired safety margin and expected cardiac event rate being detected.

(206) Accordingly, the techniques disclosed herein promote programming of an ICD or other cardiac device to a sensitivity setting that provides an acceptable safety margin for sensing cardiac events, e.g., for reliably detecting tachyarrhythmia, while avoiding oversensing of noise or other signals. When oversensing is an issue for an individual patient, the techniques disclosed herein enable a user to visualize how tachyarrhythmia detection is expected to be affected (e.g., by a change in time to detect) by a potential programming change before the programming change goes into effect. The analysis of multiple sensing electrode vector signals according to multiple sensing threshold control parameter settings may be performed efficiently without requiring each vector signal to be sensed by the ICD (or other medical device) and without having to perform time consuming operations of programming the ICD (or other medical device) to multiple different sensing control parameter settings in order to actually sense cardiac events by sensing circuit **86**

and determine actual sensed cardiac event intervals and any corresponding arrhythmia detection. The expected performance of the ICD (or other medical device) in detecting a tachyarrhythmia may be determined for multiple sensing control parameter combinations based on a single tachyarrhythmia episode, induced or spontaneous, thereby eliminating the need to perform multiple tachyarrhythmia inductions. Furthermore, while FIG. 17 is described as identifying a first sensing threshold control parameter setting and then testing sensitivity settings, any combination of two or more sensing threshold control parameters, along with one or more sensing electrode vectors, may be evaluated according to the techniques disclosed herein.

(207) It should be understood that, depending on the example, certain acts or events of any of the methods described herein can be performed in a different sequence, may be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the method). Moreover, in certain examples, acts or events may be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors, rather than sequentially. In addition, while certain aspects of this disclosure are described as being performed by a single circuit or unit for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of units or circuits associated with, for example, a medical device.

(208) In one or more examples, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored as one or more instructions or code on a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include computer-readable storage media, which corresponds to a tangible medium such as data storage media (e.g., RAM, ROM, EEPROM, flash memory, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer).

(209) Instructions may be executed by one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable logic arrays (FPLAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the term “processor,” as used herein may refer to any of the foregoing structure or any other structure suitable for implementation of the techniques described herein. Also, the techniques could be fully implemented in one or more circuits or logic elements.

(210) Thus, a medical device system has been presented in the foregoing description with reference to specific examples. It is to be understood that various aspects disclosed herein may be combined in different combinations than the specific combinations presented in the accompanying drawings. It is appreciated that various modifications to the referenced examples may be made without departing from the scope of the disclosure and the following claims.

Claims

1. A medical device system comprising: a processor configured to: receive a first cardiac electrical signal and a second cardiac electrical signal associated with a tachyarrhythmia; construct a third cardiac electrical signal from the first cardiac electrical signal and the second cardiac electrical signal; determine a first predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal according to a first setting of a plurality of settings of a sensing control parameter; determine a second predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal according to a second setting of a plurality of settings of the sensing control parameter; determine at least one acceptable setting of the sensing control parameter based on at least the first predicted time to detect the tachyarrhythmia and the second predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal; and generate an output based on the determined acceptable setting of the sensing control parameter.
2. The medical device system of claim 1, wherein the processor is further configured to: for at least

one of the first cardiac electrical signal, the second cardiac electrical signal and the constructed third cardiac electrical signal, determine a plurality of predicted times to detect the tachyarrhythmia, where each of the plurality of predicted times to detect the tachyarrhythmia is determined for a respective one of a plurality of settings of the sensing control parameter; determine that at least one of the plurality of the predicted times to detect the tachyarrhythmia is within a tachyarrhythmia detection time limit; and determine the at least one acceptable setting of the sensing control parameter by identifying at least one of the plurality of settings of the sensing control parameter for which the determined predicted time of the plurality of predicted times to detect the tachyarrhythmia is within the tachyarrhythmia detection time limit.

3. The medical device system of claim 2, wherein the processor is further configured to: determine an actual time to detect the tachyarrhythmia from one of the first cardiac electrical signal or the second cardiac electrical signal; and determine the tachyarrhythmia detection time limit by: determining a minimum time to detect the tachyarrhythmia from among the actual time to detect the tachyarrhythmia, the first predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal, and the plurality of predicted times to detect the tachyarrhythmia; and determining the tachyarrhythmia detection time limit based on the minimum time to detect the tachyarrhythmia.

4. The medical device system of claim 2 wherein the processor is further configured to: determine each of the plurality of predicted times to detect the tachyarrhythmia for the respective one of the plurality of settings of the sensing control parameter by determining each of the plurality of predicted times to detect the tachyarrhythmia for each of a plurality of sensitivity settings used to set a lowest amplitude of a cardiac event sensing threshold; and determine the at least one acceptable setting of the sensing control parameter by: determining a highest value of the plurality of sensitivity settings for which the determined predicted time to detect the tachyarrhythmia is within the tachyarrhythmia detection time limit; and determining the at least one acceptable setting of the sensing control parameter by determining a sensitivity that is a factor of the highest value of the plurality of sensitivity settings for which the determined predicted time to detect the tachyarrhythmia is within the tachyarrhythmia detection time limit.

5. The medical device system of claim 1 wherein the processor is further configured to determine the first predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal by: determining sensed cardiac events from the constructed third cardiac signal; determining sensed cardiac event intervals from the sensed cardiac events; and determining the first predicted time to detect the tachyarrhythmia based on the determined sensed cardiac event intervals.

6. The medical device system of claim 1 wherein the processor is further configured to determine the at least one acceptable setting of the sensing control parameter by determining at least one of: a sensing electrode vector; a sensitivity used to set a cardiac event sensing threshold amplitude; a percentage used to set a starting amplitude of the cardiac event sensing threshold amplitude; a percentage used to set an intermediate amplitude of the cardiac event sensing threshold amplitude; or a time interval used for making an amplitude adjustment to the cardiac event sensing threshold amplitude.

7. The medical device system of claim 1 further comprising a sensing circuit configured to sense the first cardiac electrical signal and the second cardiac electrical signal; a control circuit configured to detect the tachyarrhythmia from at least one of the first cardiac electrical signal and the second cardiac electrical signal; and a therapy delivery circuit configured to deliver a therapy in response to the control circuit detecting the tachyarrhythmia.

8. The medical device system of claim 7 further comprising: an implantable medical device comprising: a housing enclosing the sensing circuit, the control circuit and the therapy delivery circuit; and a first telemetry circuit configured to transmit the first cardiac electrical signal and the second cardiac electrical signal; and an external device comprising: a second telemetry configured to receive the first cardiac electrical signal and the second cardiac electrical signal transmitted by

the first telemetry circuit; the processor configured to receive the first cardiac electrical signal and the second cardiac electrical signal via the second telemetry circuit; and a display unit configured to display the output generated by the processor.

9. The medical device system of claim 8, wherein the processor is further configured to: for at least one of the first cardiac electrical signal, the second cardiac electrical signal and the constructed third cardiac electrical signal, determine a plurality of predicted times to detect the tachyarrhythmia, where each of the plurality of predicted times to detect the tachyarrhythmia are determined for a respective one of a plurality of sensitivity settings; generate the output by: for each one of the plurality of sensitivity settings determining a safety margin for sensing cardiac events; and determining the at least one acceptable sensing control parameter by determining an acceptable sensitivity setting based on at least the determined safety margins and the plurality of predicted times to detect the tachyarrhythmia; and the display unit is further configured to display at least a portion of the plurality of predicted times to detect the tachyarrhythmia along with a corresponding sensitivity setting and a corresponding determined safety margin.

10. A method comprising: receiving a first cardiac electrical signal and a second cardiac electrical signal associated with a tachyarrhythmia; constructing a third cardiac electrical signal from the first cardiac electrical signal and the second cardiac electrical signal; determining a first predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal according to a first setting of a plurality of settings of a sensing control parameter; determining a second predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal according to a second setting of a plurality of settings of the sensing control parameter; determining at least one acceptable setting of the sensing control parameter based on at least the first predicted time to detect the tachyarrhythmia and the second predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal; and generating an output based on the determined acceptable setting of the sensing control parameter.

11. The method of claim 10 further comprising: for at least one of the first cardiac electrical signal, the second cardiac electrical signal and the constructed third cardiac electrical signal, determining a plurality of predicted times to detect the tachyarrhythmia, where each of the plurality of predicted times to detect the tachyarrhythmia is determined for a respective one of a plurality of settings of the sensing control parameter; determining that at least one of the plurality of the predicted times to detect the tachyarrhythmia is within a tachyarrhythmia detection time limit; and determining the at least one acceptable setting of the sensing control parameter by identifying at least one of the plurality of settings of the sensing control parameter for which the determined predicted time to detect the tachyarrhythmia is within the tachyarrhythmia detection time limit.

12. The method of claim 11 further comprising: determining an actual time to detect the tachyarrhythmia from one of the first cardiac electrical signal or the second cardiac electrical signal; and determining the tachyarrhythmia detection time limit by: determining a minimum time to detect the tachyarrhythmia from among the actual time to detect the tachyarrhythmia, the first predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal, and the plurality of predicted times to detect the tachyarrhythmia; and determining the tachyarrhythmia detection time limit based on the minimum time to detect the tachyarrhythmia.

13. The method of claim 11 further comprising: determining each of the plurality of predicted times to detect the tachyarrhythmia for the respective one of the plurality of settings of the sensing control parameter comprises determining each of the plurality of predicted times to detect the tachyarrhythmia for each of a plurality of sensitivity settings used to set a lowest amplitude of a cardiac event sensing threshold; and determining the at least one acceptable setting of the sensing control parameter by: determining a highest value of the plurality of sensitivity settings for which the determined predicted time to detect the tachyarrhythmia is within the tachyarrhythmia detection time limit; and determining the at least one acceptable setting of the sensing control parameter by determining a sensitivity that is a factor of the highest value of the plurality of sensitivity settings

for which the determined predicted time to detect the tachyarrhythmia is within the tachyarrhythmia detection time limit.

14. The method of claim 10 wherein determining the first predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal comprises: determining sensed cardiac events from the constructed third cardiac signal; determining sensed cardiac event intervals from the sensed cardiac events; and determining the first predicted time to detect the tachyarrhythmia based on the determined sensed cardiac event intervals.

15. The method of claim 10 wherein determining the at least one acceptable setting of the sensing control parameter comprises determining at least one of: a sensing electrode vector; a sensitivity used to set a cardiac event sensing threshold amplitude; a percentage used to set a starting amplitude of the cardiac event sensing threshold amplitude; a percentage used to set an intermediate amplitude of the cardiac event sensing threshold amplitude; or a time interval used for making an amplitude adjustment to the cardiac event sensing threshold amplitude.

16. The method of claim 10 further comprising: sensing the first cardiac electrical signal and the second cardiac electrical signal; detecting the tachyarrhythmia from at least one of the first cardiac electrical signal and the second cardiac electrical signal; and delivering a therapy in response to detecting the tachyarrhythmia.

17. The method of claim 16 further comprising: transmitting the first cardiac electrical signal and the second cardiac electrical signal by a first telemetry circuit; receiving the first cardiac electrical signal and the second cardiac electrical signal via a second telemetry circuit; displaying by a display unit the generated output.

18. The method of claim 17 further comprising: for at least one of the first cardiac electrical signal, the second cardiac electrical signal and the constructed third cardiac electrical signal, determining a plurality of predicted times to detect the tachyarrhythmia, where each of the plurality of predicted times to detect the tachyarrhythmia are determined for a respective one of a plurality of sensitivity settings; generating the output by: for each one of the plurality of sensitivity settings, determining a safety margin for sensing cardiac events; and determining the at least one acceptable sensing control parameter by determining an acceptable sensitivity setting based on at least the determined safety margins and the plurality of predicted times to detect the tachyarrhythmia; and displaying at least a portion of the plurality of predicted times to detect the tachyarrhythmia along with a corresponding sensitivity setting and a corresponding determined safety margin.

19. A non-transitory computer readable medium storing a set of instructions that when executed by processing circuitry of a medical device system cause the medical device system to: receive a first cardiac electrical signal and a second cardiac electrical signal associated with a tachyarrhythmia; construct a third cardiac electrical signal from the first cardiac electrical signal and the second cardiac electrical signal; determine a first predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal according to a first setting of a plurality of settings of a sensing control parameter; determine a second predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal according to a second setting of a plurality of settings of the sensing control parameter; determine at least one acceptable setting of the sensing control parameter based on at least the first predicted time to detect the tachyarrhythmia and the second predicted time to detect the tachyarrhythmia from the constructed third cardiac electrical signal; generate an output based on the determined acceptable setting of the sensing control parameter; and display the generated output including the determined acceptable setting of the sensing control parameter.
